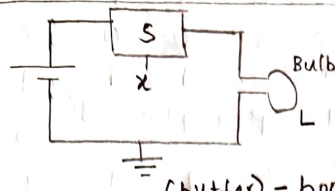
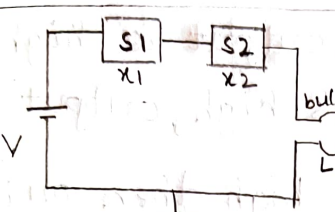
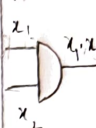
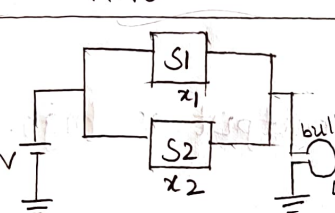
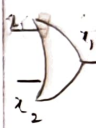
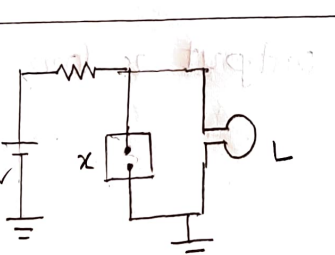
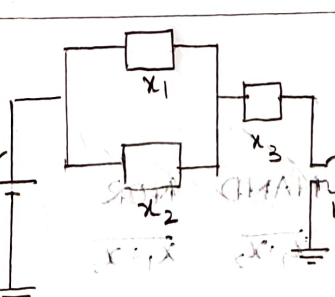
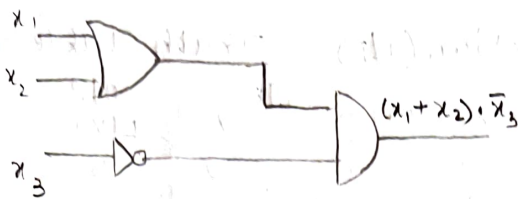


23ECE205 - Digital Electronics

	O/P = Function (I/P)	Truth Table	Symbol																																					
1.	 (buffer) - boosts the signal $x=0$ switch OFF or open  $x=1$ switch ON or close 	$L(x) = x \rightarrow$ Logical expression $L(0) = 0$ $L(1) = 1$	<table><tr><th>I/P</th><th>O/P</th></tr><tr><th>x</th><th>$L(x)$</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr></table>	I/P	O/P	x	$L(x)$	0	0	1	1																													
I/P	O/P																																							
x	$L(x)$																																							
0	0																																							
1	1																																							
2.	 AND	Series connection = logical AND operation $L(x_1, x_2) = x_1 \cdot x_2$	<table><tr><th>I/P</th><th>O/P</th></tr><tr><th>x_1</th><th>x_2</th><th>$L(x_1, x_2)$</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1 (ON)</td></tr></table> 	I/P	O/P	x_1	x_2	$L(x_1, x_2)$	0	0	0	0	1	0	1	0	0	1	1	1 (ON)																				
I/P	O/P																																							
x_1	x_2	$L(x_1, x_2)$																																						
0	0	0																																						
0	1	0																																						
1	0	0																																						
1	1	1 (ON)																																						
3.	 OR	Parallel connection = logical OR operation $L(x_1, x_2) = x_1 + x_2$	<table><tr><th>I/P</th><th>O/P</th></tr><tr><th>x_1</th><th>x_2</th><th>$x_1 + x_2$</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1 (ON)</td></tr><tr><td>1</td><td>0</td><td>1 (ON)</td></tr><tr><td>1</td><td>1</td><td>1 (ON)</td></tr></table> 	I/P	O/P	x_1	x_2	$x_1 + x_2$	0	0	0	0	1	1 (ON)	1	0	1 (ON)	1	1	1 (ON)																				
I/P	O/P																																							
x_1	x_2	$x_1 + x_2$																																						
0	0	0																																						
0	1	1 (ON)																																						
1	0	1 (ON)																																						
1	1	1 (ON)																																						
4.	 NOT	Logical inversion $L(x) = \bar{x}$ $= \sim x$ $= x'$	<table><tr><th>x</th><th>$L(x)$</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table> 	x	$L(x)$	0	1	1	0																															
x	$L(x)$																																							
0	1																																							
1	0																																							
5.		$L(x_1, x_2, x_3)$ $= (x_1 + x_2) \cdot x_3$	<table><tr><th>x_1</th><th>x_2</th><th>x_3</th><th>$L(x_1, x_2, x_3)$</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> 	x_1	x_2	x_3	$L(x_1, x_2, x_3)$	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	0	0	1	0	1	1	1	1	0	0	1	1	1	1	
x_1	x_2	x_3	$L(x_1, x_2, x_3)$																																					
0	0	0	0																																					
0	0	1	0																																					
0	1	0	0																																					
0	1	1	1																																					
1	0	0	0																																					
1	0	1	1																																					
1	1	0	0																																					
1	1	1	1																																					

Ex:-

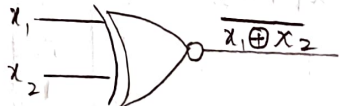


I/P			O/P		
x_1	x_2	x_3	$x_1 + x_2$	$\bar{x}_3$	$(x_1 + x_2) \cdot \bar{x}_3$
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

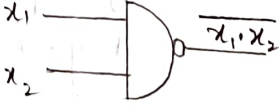
1. AND:- If both/all inputs are high, output is high.
2. OR:- If either/anyone of input is high, output is high.
3. EX-OR:- If odd no. of inputs are high then output is high.



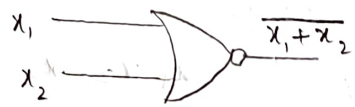
4. EX-NOR:- If both inputs are same output is high.



5. NAND:- If both inputs are high, output is low



6. NOR:- If any input is high then output is low.

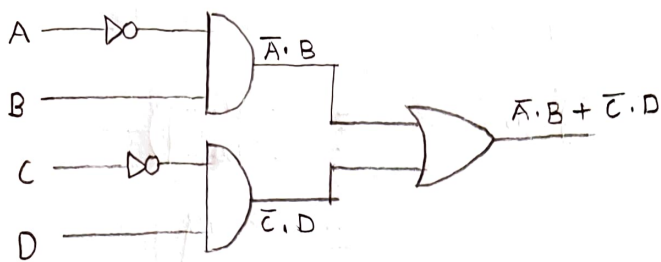


x_1	x_2	AND $x_1 \cdot x_2$	OR $x_1 + x_2$	EX-OR $x_1 \oplus x_2$	EX-NOR $\overline{x_1 \oplus x_2}$	NAND $\overline{x_1 \cdot x_2}$	NOR $\overline{x_1 + x_2}$
0	0	0	0	0	1	1	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	1	0	0

Analysis - First diagram
 Synthesis - Last is diagram.

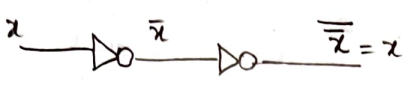
} Differences.

Ex:- $\bar{A} \cdot B + \bar{C} \cdot D$

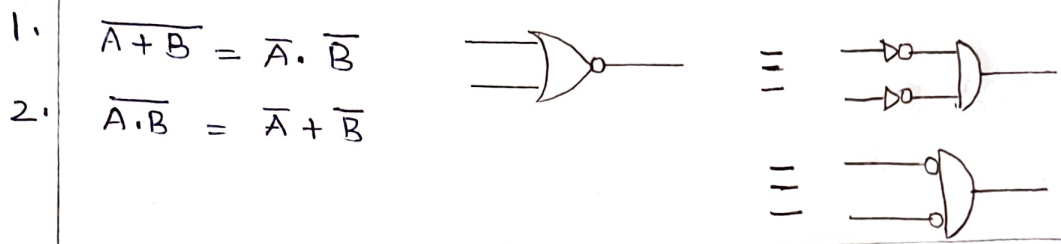


A	B	C	D	$\bar{A}$	$\bar{A} \cdot B$	$\bar{C}$	$\bar{C} \cdot D$	$\bar{A} \cdot B + \bar{C} \cdot D$
0	0	0	0	1	0	1	0	0
0	0	0	1	1	0	1	1	1
0	0	1	0	1	0	0	0	0
0	0	1	1	1	0	0	0	0
0	1	0	0	1	1	1	0	1
0	1	0	1	1	1	1	1	1
0	1	1	0	1	1	0	0	1
0	1	1	1	1	1	0	0	1
1	0	0	0	0	0	1	0	0
1	0	0	1	0	0	1	1	1
1	0	1	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0
1	1	0	0	0	0	1	0	0
1	1	0	1	0	0	1	1	1
1	1	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0

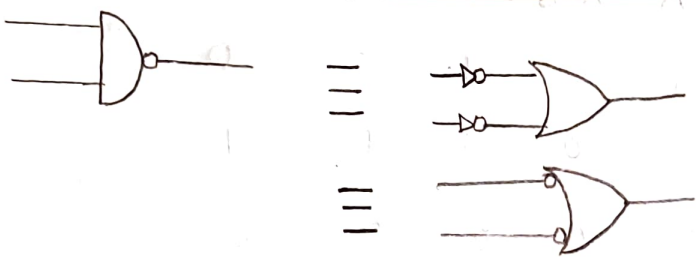
* Laws



Demorgan's Law



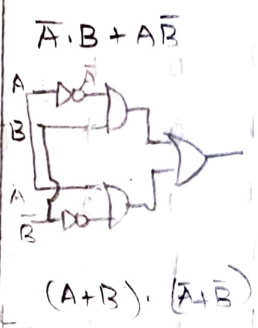
NOR = Bubbled AND gate.



NAND = Bubbled OR gate.

	Using NAND	using NOR
NOT	$\overline{x \cdot x} = \bar{x}$	$\overline{x + x} = \bar{x}$
AND	$x \cdot x = x$	
OR		$x + x = x$
EXOR	$A \oplus B = \bar{A} \cdot B + A \cdot \bar{B}$	$A \oplus B = (A+B) \cdot (\bar{A} + \bar{B})$

$$\begin{aligned} \overline{x \cdot x} &= \bar{x} + \bar{x} \\ &= \bar{x} \\ \overline{x + x} &= \bar{x} \cdot \bar{x} \\ &= \bar{x} \end{aligned}$$



* Laws of Boolean Algebra (laws related to 0 or 1)

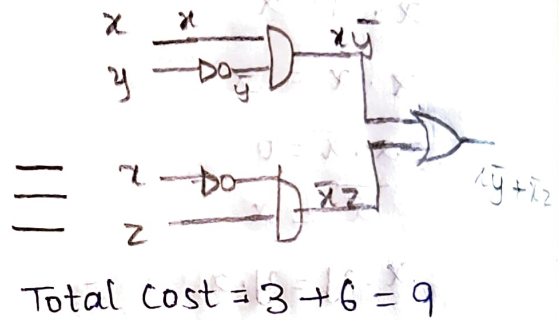
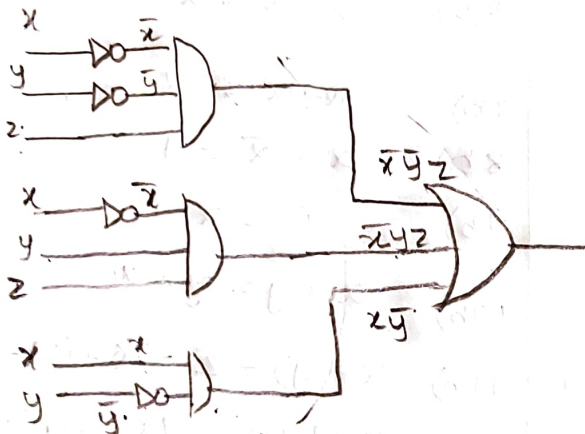
AND Gate rules	OR Gate rules.
1a) $0 \cdot 0 = 0$	1b) $1 + 1 = 1$
2a) $1 \cdot 1 = 1$	2b) $0 + 0 = 0$
3a) $0 \cdot 1 = 1 \cdot 0 = 0$	3b) $1 + 0 = 0 + 1 = 1$
4a) if $x = 0$ then $\bar{x} = 1$	4b) if $x = 1$ then $\bar{x} = 0$
5a) $x \cdot 0 = 0$	5b) $x + 1 = 1$
6a) $x \cdot 1 = x$	6b) $x + 0 = x$
7a) $x \cdot x = x$	7b) $x + x = x$
8a) $x \cdot \bar{x} = 0$	8b) $x + \bar{x} = 1$
9a) $\overline{\bar{x}} = x$	9b) $\overline{\overline{x}} = x$
10a) $x \cdot y = y \cdot x$ (commutative law)	10b) $x + y = y + x$
11a) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ (associative law)	11b) $(x + y) + z = x + (y + z)$
12a) $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ (distributive law)	12b) $x + (y \cdot z) = (x + y) \cdot (x + z)$
13a) $x + x \cdot y = x$ (absorption law)	13b) $x \cdot (x + y) = x$
14a) $x \cdot y + x \cdot \bar{y} = x$	14b) $(x + y) \cdot (x + \bar{y}) = x$
15a) $\overline{x \cdot y} = \bar{x} + \bar{y}$ (De Morgan's law)	15b) $\overline{x + y} = \bar{x} \cdot \bar{y}$
16a) $x + \bar{x} \cdot y = x + y$	16b) $x \cdot (\bar{x} + y) = x \cdot y$
17a) $x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$ (consensus theorem)	17b) $(x + y) \cdot (y + z) \cdot (\bar{x} + z) = (x + y) \cdot (\bar{x} + z)$

Dual of an logical expression is obtained by replacing $\cdot$ with $+$ & vice versa & 0 with 1 & vice versa. — (Law of Duality)

$$\begin{aligned}
 x + x \cdot y &= x \Rightarrow 1 \cdot x + x \cdot y = x \text{ (using } x \cdot 1 = x) \\
 &= x \cdot (1 + y) \text{ (using distributive law)} \\
 &= x \cdot 1 \quad 1 + x = 1 \\
 &= x
 \end{aligned}$$

$$\begin{aligned}
 x + \bar{x} \cdot y &= (x + \bar{x}) \cdot (x + y) \quad (\text{using distributive law}) \\
 &= 1 \cdot (x + y) \quad (1 \cdot x = x) \\
 &= x + y
 \end{aligned}$$

$$\begin{aligned}
 \bar{x} \cdot \bar{y} \cdot z + \bar{x} y z + x \bar{y} &= x \bar{y} + \bar{x} z \quad (\text{Taking } \bar{x} z \text{ common}) \\
 &= \bar{x} z (\bar{y} + y) + x \bar{y} \quad (\text{using } x + \bar{x} = 1) \\
 &= \bar{x} z \cdot 1 + x \bar{y} \quad (\text{using } x \cdot \bar{x} = 1) \\
 &= \bar{x} z + x \bar{y}
 \end{aligned}$$



$$\text{LHS (Total cost} = 4 + 11 = 15) \quad \text{RHS}$$

Cost of a given logical circuit is ~~total~~ no. of gates + Total no. of inputs.

Assuming inputs are available in true & complemented form is free of cost.

17a) Consensus theorem.

$$x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + \bar{x} \cdot z$$

$$\begin{aligned}
 \text{L.H.S} &= x \cdot y + y \cdot z + \bar{x} \cdot z = x \cdot y + y \cdot z \cdot 1 + \bar{x} \cdot z \\
 &= x \cdot y + y \cdot z (x + \bar{x}) + \bar{x} \cdot z \\
 &= (x \cdot y + x \cdot y \cdot z) + (\bar{x} \cdot y \cdot z + \bar{x} \cdot z) \\
 &= x \cdot y (1 + z) + \bar{x} \cdot z (1 + y) \\
 &= x \cdot y + \bar{x} \cdot z
 \end{aligned}$$

Ex:- $f = x + yz$. Find f'

$$\begin{aligned} f' &= \overline{x + yz} \\ &= \overline{x} \cdot \overline{yz} \\ &= \overline{x} \cdot (\overline{y} + \overline{z}) \end{aligned}$$

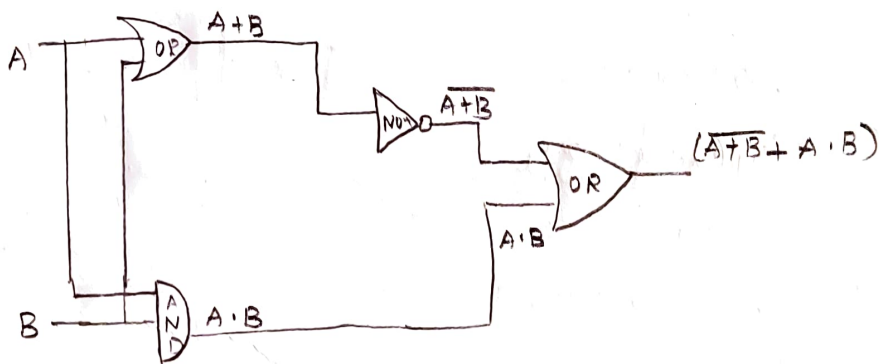
Q. $f = x + (\overline{y}z) + (\overline{x} + yz)$

$$\begin{aligned} f' &= \overline{(x + \overline{y}z) + (\overline{x} + yz)} \\ &= \overline{(x + \overline{y}z)} \cdot \overline{(\overline{x} + yz)} \\ &= (\overline{x} \cdot \overline{\overline{y}z}) \cdot (\overline{\overline{x}} \cdot \overline{yz}) \\ &= (\overline{x} \cdot (\overline{\overline{y}} + \overline{\overline{z}})) \cdot (x \cdot (\overline{y} + \overline{z})) \\ &= [\overline{x} \cdot (y + \overline{z})] \cdot [x \cdot (\overline{y} + \overline{z})] \end{aligned}$$

17/7/25

Assignment

1. Write Boolean Expression for the following circuit. Draw the Timing diagram for inputs A & B



$$(\overline{A+B} + A.B) = \overline{A} \cdot \overline{B} + A.B$$

2. Use algebraic transformation to prove or disprove that the two circuits given in Fig.1 implement the same function (do not use truth table or kmap)

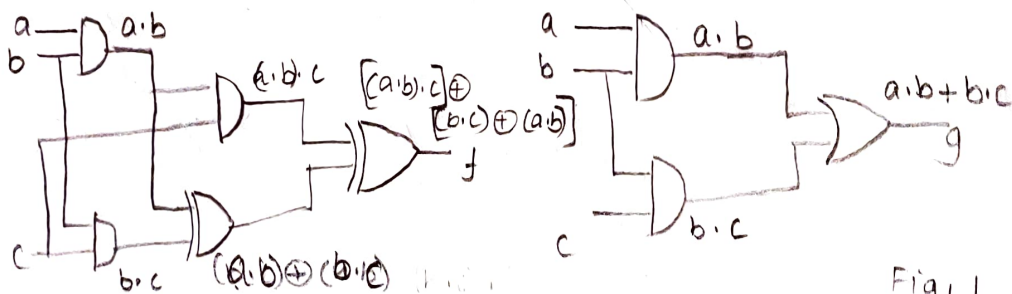


Fig. 1

$$f = [(a \cdot b) \cdot c] \oplus [(b \cdot c) \oplus (a \cdot b)] \quad g = a \cdot b + b \cdot c$$

$$= [(a \cdot b) \cdot (b \cdot c)] \oplus [(b \cdot c) \oplus (a \cdot b)]$$

$$a \cdot b = x$$

$$b \cdot c = y$$

$$= [x \cdot y] \oplus [y \oplus x]$$

$$= [x \cdot y] \oplus [\bar{x}y + \bar{y}x]$$

$$= (\bar{x} \cdot \bar{y}) [\bar{x}y + \bar{y}x] + (\bar{x}y + \bar{y}x) (x \cdot y)$$

$$= (x + y) [\bar{x}y + \bar{y}x] + [\bar{x}y \cdot \bar{y}x] (x \cdot y)$$

$$= [\bar{x} \cdot \bar{x}y + \bar{y}x \cdot \bar{x} + \bar{x}y \cdot \bar{y}x + \bar{y}y \cdot x] + [(x + y) \cdot (y + \bar{x})] (x \cdot y)$$

$$= [\bar{x}y + \bar{y}x] + [xy + \bar{x}x + \bar{y}y + \bar{x}y] (x \cdot y)$$

$$= [\bar{x}y + \bar{y}x] + [xxyy + x\bar{x}y\bar{y}]$$

$$= (\bar{x}y + \bar{y}x) + xy$$

$$= xy + \bar{x}y + \bar{y}x$$

$$= y(x + \bar{x}) + x\bar{y}$$

$$= y +$$

3. Write Truth table for both circuits & verify their equality.

a	b	c	$a \cdot b$	$b \cdot c$	$(a \cdot b) \cdot c$	$(a \cdot b) \oplus (b \cdot c)$	$(a \cdot b) \cdot c \oplus [(b \cdot c) \oplus (a \cdot b)]$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	1	0	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	1	1	0	0

equivalent

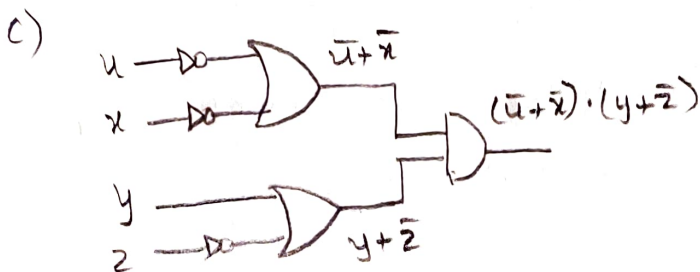
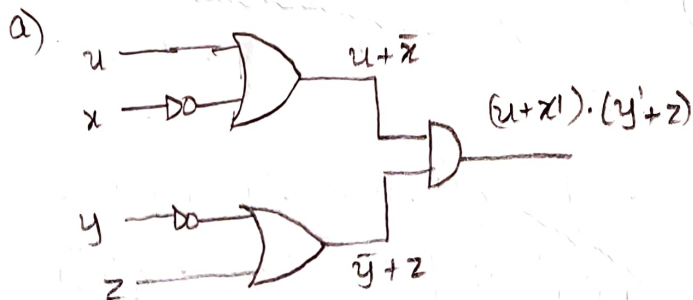
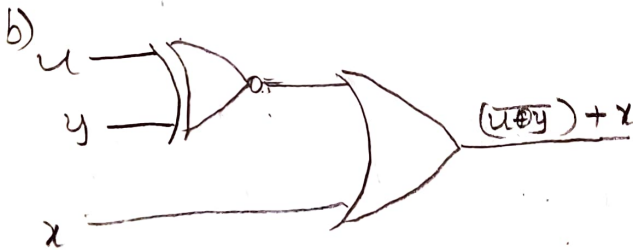
4.

a) $(u+x')(y'+z)$

b) $(u \oplus y)' + x$

c) $(u'+x')(y+z')$

Draw logic diagrams to implement the following boolean expression.



7. Simplify following boolean expressions to min. no. of literals.

- (a) $ABC + A'B + ABC'$ (b) $x'yz + xz$
 (c) $(x+y)'(x'+y')$ (d) $xy + x(wz + wz')$
 (e) $(Bc' + A'D)(AB' + CD')$ (f) $(a' + c')(a + b' + c')$

$$\begin{aligned} \text{a) } ABC + A'B + ABC' &= AB(C + C') + A'B && (x + x' = 1) \\ &= AB + A'B = B(A + A') = B \end{aligned}$$

$$\begin{aligned} \text{b) } x'yz + xz &= z(x'y + x) \\ &= z(x + y) \\ &= xz + xy && (x + \bar{x}y = x + y) \end{aligned}$$

$$\begin{aligned} \text{c) } (x+y)'(x'+y') &= (\bar{x} \cdot \bar{y}) \cdot (\bar{x}' + \bar{y}) \\ &= \bar{x}\bar{y} \cdot \bar{x} + \bar{x}\bar{y} \cdot \bar{y} = \bar{x}\bar{y} + \bar{x}\bar{y} \\ &= \bar{x}\bar{y} \end{aligned}$$

$$\begin{aligned} \text{d) } xy + x(wz + wz') &= xy + xw(z + z') \\ &= xy + xw(1) \\ &= x(y + w) \end{aligned}$$

$$\begin{aligned} \text{e) } (Bc' + A'D)(AB' + CD') &= \cancel{AB'Bc'} + \cancel{Bc'CD'} && (A \cdot \bar{A} = 0) \\ &\quad + \cancel{A'DAB'} + \cancel{A'DCD'} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{f) } (a' + c')(a + b' + c') &= aa' + a'b' + a'c' + ac' + b'c' + c'c' \\ &= 0 + a'b' + a'c' + ac' + b'c' + c' \\ &= b'(a' + c') + c'(a' + a) + c' && (x + 1 = 1) \\ &= b'(a' + c') + c' + c' \\ &= a'b' + b'c' + c' \end{aligned}$$

25/7/25

* **Minterm**:- Minterm is a product term where each input present either in true or complement form.

Row No	x_1	x_2	Minterm
0	0	0	$M_0 = \bar{x}_1 \cdot \bar{x}_2$
1	0	1	$M_1 = \bar{x}_1 \cdot x_2$
2	1	0	$M_2 = x_1 \cdot \bar{x}_2$
3	1	1	$M_3 = x_1 \cdot x_2$

Ex:-

Row No	x_1	x_2	$Y_{(XOR)}$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$$Y = \bar{x}_1 \cdot x_2 + x_1 \cdot \bar{x}_2$$

$$= m_1 + m_2$$

$$= \sum m(1, 2)$$

$$\text{Maxterm} = M_0 \cdot M_3 = (x_1 + x_2) \cdot (\bar{x}_1 + \bar{x}_2)$$

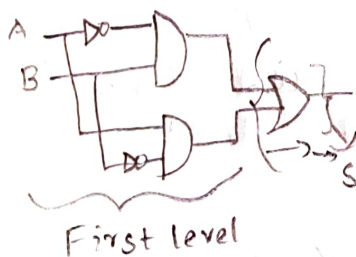
SOP - sum of product terms. Ex:- $\bar{A}B + A\bar{B}$

↓

→ More specifically it is Canonical SOP Expression.

SOP diagram:-

(because each term is a minterm)
& all variable should be present to be canonical.



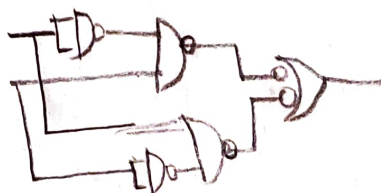
⇒ Two level logic diagram.

Second level

First level

→ Every SOP diagram will be in 2 level diagram
1st level - AND, 2nd level - OR → These can be easily implemented using universal gates.

XNOR using

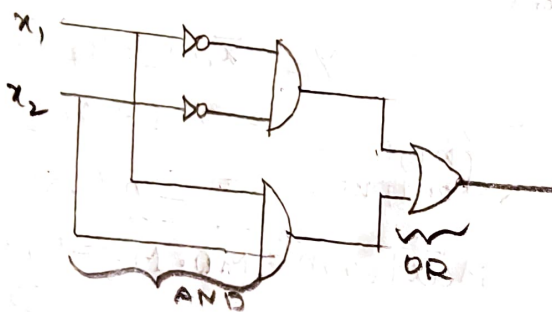


* Minterm using X-NOR implement with normal & universal gates.

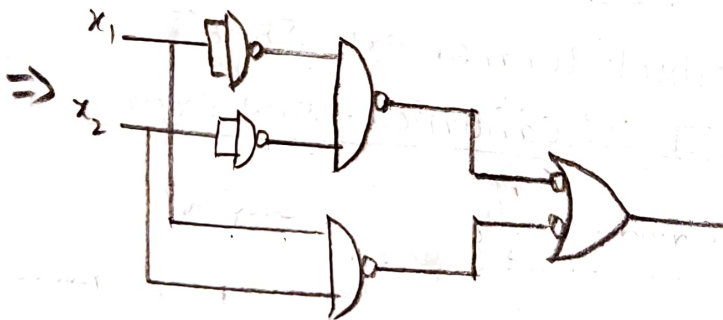
Row No	x_1	x_2	X-NOR $\overline{x_1 \oplus x_2}$ minterm	Maxterm
0	0	0	$\checkmark 1 = m_0$	$M_0 = x_1 + x_2$
1	0	1	$0 = m_1$	$M_1 = x_1 + \overline{x_2}$ ✓
2	1	0	$0 = m_2$	$M_2 = \overline{x_1} + x_2$ ✓
3	1	1	$\checkmark 1 = m_3$	$M_3 = \overline{x_1} + \overline{x_2}$

Minterm = $m_0 + m_3 = \sum m(0, 3) \rightarrow$ minterm list
 $= \overline{x_1} \cdot \overline{x_2} + x_1 \cdot x_2$

corresponding to truth table.



AND-OR gates
 cost = 3 + 6 = 9



NAND-NAND gates

cost = 8 + 5
 = 13

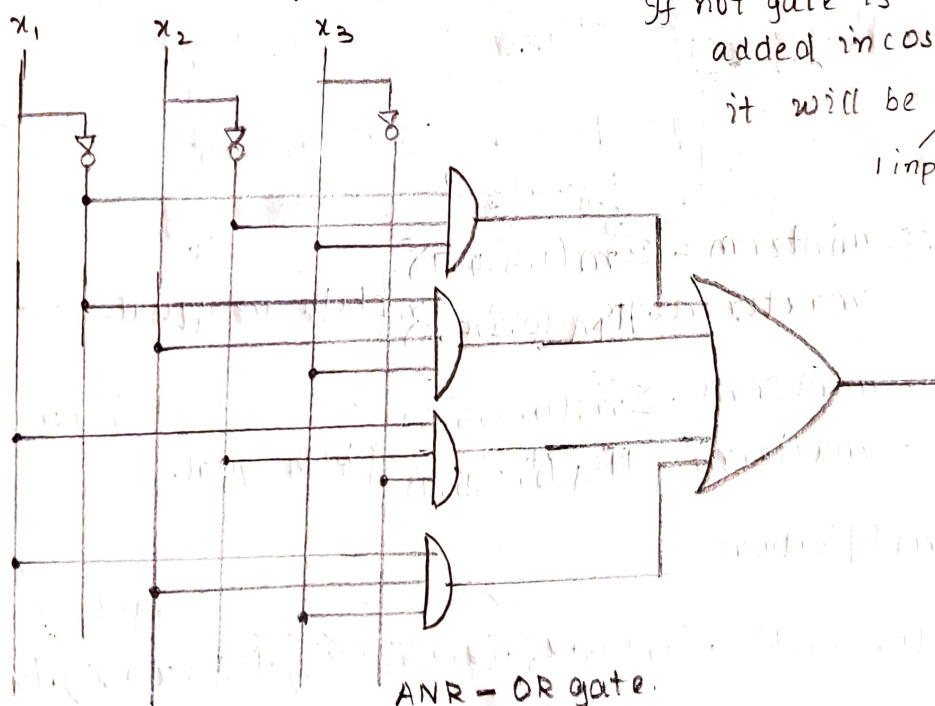
using NAND

Row No	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \bar{x}_1 \cdot \bar{x}_2 \cdot \bar{x}_3$	$M_0 = \bar{m}_0 = (\bar{x}_1 \cdot \bar{x}_2 \cdot \bar{x}_3) = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \bar{x}_1 \cdot \bar{x}_2 \cdot x_3$	$M_1 = \bar{m}_1 = x_1 + x_2 + \bar{x}_3$
2	0	1	0	$m_2 = \bar{x}_1 \cdot x_2 \cdot \bar{x}_3$	$M_2 = \bar{m}_2 = x_1 + \bar{x}_2 + x_3$
3	0	1	1	$m_3 = \bar{x}_1 \cdot x_2 \cdot x_3$	$M_3 = \bar{m}_3 = x_1 + \bar{x}_2 + \bar{x}_3$
4	1	0	0	$m_4 = x_1 \cdot \bar{x}_2 \cdot \bar{x}_3$	$M_4 = \bar{m}_4 = \bar{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1 \cdot \bar{x}_2 \cdot x_3$	$M_5 = \bar{m}_5 = \bar{x}_1 + x_2 + \bar{x}_3$
6	1	1	0	$m_6 = x_1 \cdot x_2 \cdot \bar{x}_3$	$M_6 = \bar{m}_6 = \bar{x}_1 + \bar{x}_2 + x_3$
7	1	1	1	$m_7 = x_1 \cdot x_2 \cdot x_3$	$M_7 = \bar{m}_7 = \bar{x}_1 + \bar{x}_2 + \bar{x}_3$

Ex:- $m_1 + m_3 + m_4 + m_7 = \sum m(1, 3, 4, 7)$

Logical Expression = $\bar{x}_1 \cdot \bar{x}_2 \cdot x_3 + \bar{x}_1 \cdot x_2 \cdot x_3 + x_1 \cdot \bar{x}_2 \cdot \bar{x}_3 + x_1 \cdot x_2 \cdot x_3$

→ Canonical SOP expression.



If not gate is added in cost then it will be +2
input 1 gate

Cost = 5 gates + 4(3) + 4(inputs) = 5 + 12 + 4 = 21

NAND gates = 3 NOT + 4 AND + 1 OR = 8

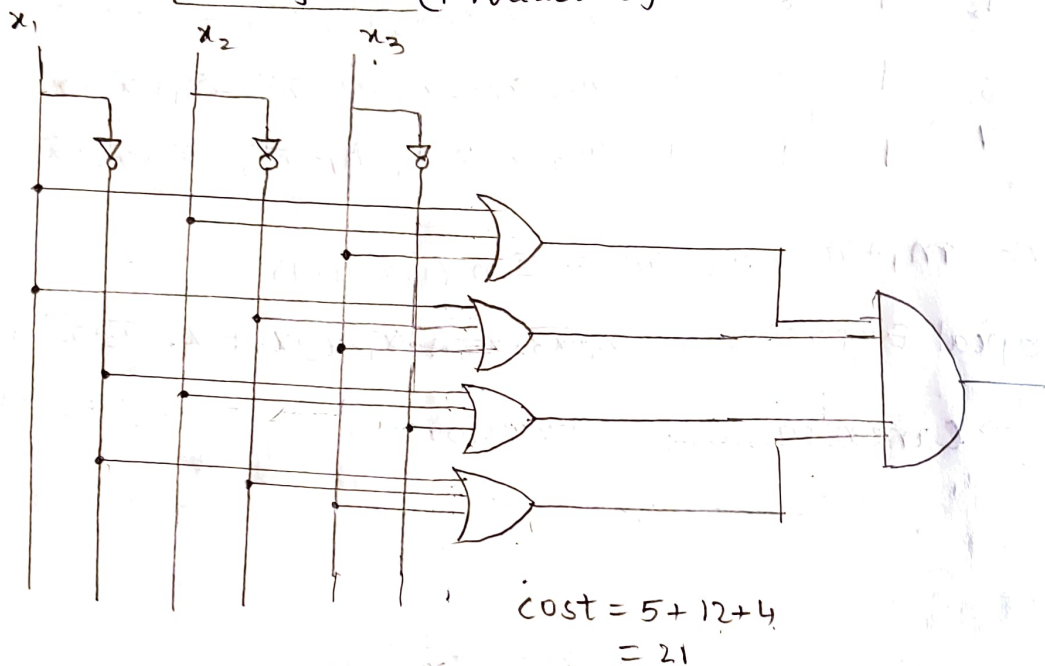
→ $\bar{x}_1 \cdot \bar{x}_2 \cdot x_3 + \bar{x}_1 \cdot x_2 \cdot x_3 + x_1 \cdot \bar{x}_2 \cdot \bar{x}_3 + x_1 \cdot x_2 \cdot x_3$

= $x_2 \cdot x_3 (x_1 + \bar{x}_1) + \bar{x}_2 (\bar{x}_1 \cdot x_3 + x_1 \cdot x_3) = x_2 \cdot x_3 + \bar{x}_2 (x_1 \oplus x_3)$

→ SOP but not canonical

Maxterm :- It is a sum term in which all the input variables are present in true or complement form.

Ex:- $M(0, 2, 5, 6) = M_0 \cdot M_2 \cdot M_5 \cdot M_6 = \Pi_M(0, 2, 5, 6)$
 $= (x_1 + x_2 + \bar{x}_3) \cdot (x_1 + \bar{x}_2 + \bar{x}_3) \cdot (\bar{x}_1 + x_2 + \bar{x}_3) \cdot (\bar{x}_1 + \bar{x}_2 + \bar{x}_3)$
 $\hookrightarrow$ ^{canonical} POS form (Product of sums)



Ex:- $f = \text{minterm} = \Sigma m(1, 3, 4, 7)$
 $\text{maxterm} = \Pi_M(0, 2, 5, 6)$ } both are equal.

$f' = \text{minterm} = \Sigma m(0, 2, 5, 6)$
 $= \text{maxterm} = \Pi_M(1, 3, 4, 7)$ } both equal.

Simplification:-

$$\begin{aligned} (a+b) \cdot (a+b) &= a & (x+y) \cdot (x+z) &= x + (y \cdot z) \\ M(0, 2, 5, 6) &= (x_1 + x_2 + x_3) \cdot (x_1 + \bar{x}_2 + x_3) \cdot (\bar{x}_1 + x_2 + \bar{x}_3) \cdot (\bar{x}_1 + \bar{x}_2 + \bar{x}_3) \\ &= (x_1 + x_3) \cdot \left[(x_1) + (x_2 + \bar{x}_3)(\bar{x}_2 + x_3) \right] \\ &= (x_1 + x_3) \left[\bar{x}_1 + \cancel{x_1 x_2} + x_2 x_3 + \bar{x}_2 \bar{x}_3 + \cancel{x_3 \bar{x}_2} \right] \\ &= \cancel{x_1 x_1} + \cancel{x_1 x_2} + x_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + \bar{x}_1 x_3 + x_2 x_3 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \\ &= \bar{x}_1 x_3 + x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 \\ &= \bar{x}_1 x_3 + x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 \end{aligned}$$

→ POS logical diagram is a 2-level circuit & can easily be converted NOR-NOR gates.

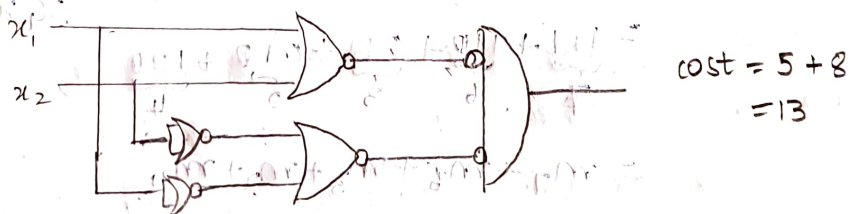
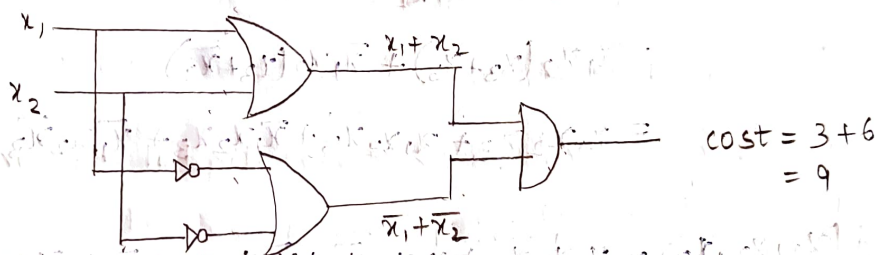
Maxterm for X-OR

Row No	x_1	x_2	XOR
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$$\rightarrow M_0 = x_1 + x_2$$

$$\rightarrow M_3 = \bar{x}_1 + \bar{x}_2$$

$$\text{Maxterm} = \Pi M(0, 3) = M_0 \cdot M_3 = (x_1 + x_2) \cdot (\bar{x}_1 + \bar{x}_2)$$



NOR-NOR gate

Q. Design 3-Input XOR gate (Minterm & Maxterm)

Row No	A	B	C	XOR $A \oplus B \oplus C$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

$$\rightarrow M_0$$

$$\rightarrow m_1$$

$$\rightarrow m_2$$

$$\rightarrow M_3$$

$$\rightarrow m_4$$

$$\rightarrow M_5$$

$$\rightarrow M_6$$

$$\rightarrow m_7$$

$$\text{Minterm} = \sum m(1, 2, 4, 7)$$

$$\text{Maxterm} = \Pi M(0, 3, 5, 6)$$

Q. $f(x_1, x_2, x_3) = x_2 + x_1 \bar{x}_3$. Write minterm for following function.

$$f(x_1, x_2, x_3) = x_2 + x_1 \bar{x}_3$$

$$x_1 \bar{x}_3 = x_1 \cdot 1 \cdot \bar{x}_3$$

$$= x_1 \cdot (x_2 + \bar{x}_2) \cdot \bar{x}_3$$

$$= x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3$$

$$x_2 = 1 \cdot x_2 = (x_1 + \bar{x}_1) \cdot x_2$$

$$= x_1 x_2 + \bar{x}_1 x_2$$

$$= x_1 x_2 \cdot 1 + \bar{x}_1 x_2 \cdot 1$$

$$= x_1 x_2 (x_3 + \bar{x}_3) + \bar{x}_1 x_2 (x_3 + \bar{x}_3)$$

$$= x_1 x_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3$$

$$\therefore f(x_1, x_2, x_3) = x_1 x_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3$$

$$= \underbrace{111}_7 + \underbrace{110}_6 + \underbrace{011}_3 + \underbrace{010}_2 + \underbrace{100}_4$$

$$= m_7 + m_6 + m_3 + m_2 + m_4$$

$$= \sum m(7, 6, 3, 2, 4) \rightarrow \text{minterm (SOP)}$$

$$x_2 + x_1 \bar{x}_3 = (x_2 + x_1) \cdot (x_2 + \bar{x}_3)$$

$$= (x_1 + x_2 + 0) \cdot (0 + x_2 + \bar{x}_3)$$

$$= (x_1 + x_2 + x_3 \bar{x}_3) \cdot (x_1 \bar{x}_1 + x_2 + \bar{x}_3)$$

$$= (x_1 + x_2 + x_3) \cdot (x_1 + x_2 + \bar{x}_3) \cdot (x_1 + x_2 + \bar{x}_3)$$

$$= (x_1 + x_2 + x_3) \cdot (x_1 + x_2 + \bar{x}_3) \cdot (\bar{x}_1 + x_2 + \bar{x}_3)$$

$$= \begin{matrix} 0 & 0 & 0 & 0 & 0 & 1 \\ M_0 & M_1 & M_2 & M_3 & M_4 & M_5 \end{matrix}$$

$$\begin{aligned}
 9. \quad f &= ab + bc + ca \\
 &= ab \cdot 1 + 1 \cdot bc + ca \cdot 1 \\
 &= ab(c + \bar{c}) + (a + \bar{a})bc + ca(b + \bar{b}) \\
 &= abc + ab\bar{c} + abc + \bar{a}bc + abc + a\bar{b}c \\
 &= abc + ab\bar{c} + \bar{a}bc + a\bar{b}c \\
 &\quad \quad \quad \begin{array}{cccc} 111 & 110 & 011 & 101 \end{array} \\
 &= m_7 + m_6 + m_3 + m_5 \\
 &= \sum m(3, 5, 6, 7) \rightarrow \text{SOP}
 \end{aligned}$$

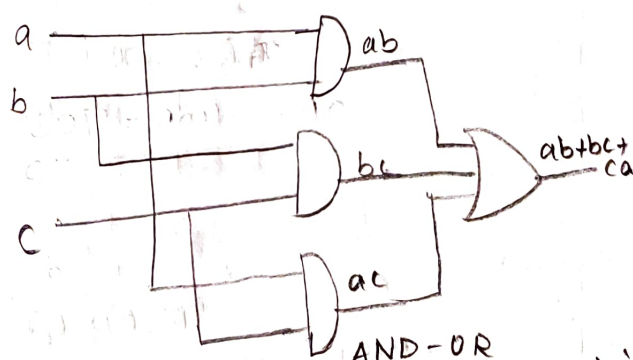
$$\begin{aligned}
 f &= ab + bc + ca \\
 &= \pi M(0, 1, 2, 4) \rightarrow \text{POS}
 \end{aligned}$$

Q. Write a function f with 3 inputs a, b, c where $f=1$ when majority of inputs are 1. Find cost of NAND gate implementation.

R.No	a	b	c	f
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

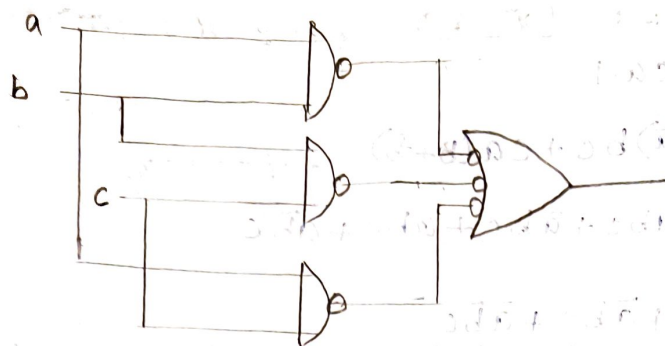
$$\begin{aligned}
 \text{SOP} &= \sum m(3, 5, 6, 7) \\
 &= m_3 + m_5 + m_6 + m_7
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow m_3 = \bar{a}bc + \bar{a}\bar{b}c + a\bar{b}c + abc \\
 &= bc(\bar{a} + \bar{a}) + a\bar{b}c + abc \\
 &= bc + a\bar{b}c + abc \\
 &= c(\bar{b} + a\bar{b}) + abc \\
 &= c(a + b) + abc \\
 &= ac + bc + abc \\
 &= a(c + bc) + bc \\
 &= a(b + c) + bc \\
 &= ab + ac + bc
 \end{aligned}$$



$$\text{Cost} = 4 + 9 = 13$$

$$\therefore [x + \bar{x}y = x + y]$$



NAND - NAND

$$\text{Cost} = 4 + 9$$

$$= 13$$

$$\begin{aligned} \text{POS} &= \Pi M(0, 1, 2, 4) = M_0 \cdot M_1 \cdot M_2 \cdot M_4 = (a+b+c) \cdot (a+b+\bar{c}) \cdot (a+\bar{b}+c) \cdot (\bar{a}+\bar{b}+\bar{c}) \\ &= (a+b) \cdot (a\bar{c} + ab + ac + \bar{a}\bar{b} + \bar{b}c + c\bar{a} + cb + cc) \\ &= (a+b) \cdot (ab + ac + \bar{a}\bar{b} + \bar{b}c + c\bar{a} + cb + c) \\ &= ab + ac + 0 + \bar{a}\bar{b}c + abc + a\bar{b}c + \bar{a}b\bar{c} + bc + bc \\ &= ab + ac + 0 + abc + \bar{a}\bar{b}c + bc \\ &= ab + ac + 0 + bc + bc = ab + bc + ac \end{aligned}$$

- Q. A circuit that controls a given digital system has 3 inputs x_1, x_2, x_3 it has to recognize 3 different conditions
- a) True if x_3 is T & either x_1 is T or x_2 is F.
 - condition b) is True if x_1 is T & either x_2 or x_3 is F
 - condition c) is True if x_2 is T & either x_1 is T or x_3 is F.
- The control circuit must produce output of 1 if atleast 2 of the condition a, b, c are True. Design the simplest circuit to do this.

Row No	x_1	x_2	x_3	a	b	c	f
0	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0
2	0	1	0	0	0	1	0
3	0	1	1	0	0	0	0
4	1	0	0	0	1	0	0
5	1	0	1	1	1	0	1
6	1	1	0	0	1	1	1
7	1	1	1	1	0	1	1

$$\begin{aligned} \text{SOP} &= m_5 + m_6 + m_7 \\ &= x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3 \\ &= x_1 x_3 (\bar{x}_2 + x_2) + x_1 x_2 \bar{x}_3 \\ &= x_1 x_3 + x_1 x_2 \bar{x}_3 \\ &= x_1 (x_3 + x_2 \bar{x}_3) \\ &= x_1 (x_3 + x_2) \end{aligned}$$

→ m_5

→ m_6

→ m_7

$$a = x_3 \cdot (x_1 + \bar{x}_2), b = x_1 \cdot (\bar{x}_2 + \bar{x}_3), c = x_2 \cdot (x_1 + \bar{x}_3)$$

$$f = a \cdot b + b \cdot c + c \cdot a + a \cdot b \cdot c$$

$$= ab(1+c) + bc + ca$$

$$= ab + bc + ca$$

$$f = [x_3 \cdot (x_1 + \bar{x}_2)] \cdot [x_1 \cdot (\bar{x}_2 + \bar{x}_3)] + [x_1 \cdot (\bar{x}_2 + \bar{x}_3)] \cdot [x_2 \cdot (x_1 + \bar{x}_3)]$$

$$+ [x_2 \cdot (x_1 + \bar{x}_3)] \cdot [x_3 \cdot (x_1 + \bar{x}_2)]$$

$$= [x_1 x_3 + \bar{x}_2 x_3] [x_1 \bar{x}_2 + x_1 \bar{x}_3] + [x_1 \bar{x}_2 + x_1 \bar{x}_3] [x_1 x_2 + x_2 \bar{x}_3]$$

$$+ [x_1 x_2 + x_2 \bar{x}_3] [x_1 x_3 + \bar{x}_2 x_3]$$

$$= x_1 x_3 x_1 \bar{x}_2 + x_1 x_3 \bar{x}_2 x_1 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_2 x_1 \bar{x}_3 + x_1 \bar{x}_2 x_2 x_1 \bar{x}_3 + x_1 \bar{x}_2 x_1 x_2 + x_1 \bar{x}_2 \bar{x}_2 x_2 \bar{x}_3$$

$$+ x_1 \bar{x}_3 x_1 x_2 + x_1 \bar{x}_3 \bar{x}_2 x_2 \bar{x}_3 + x_1 x_2 x_1 x_3 + x_1 x_2 \bar{x}_2 x_3 + x_2 \bar{x}_3 x_1 x_3 + x_2 \bar{x}_3 \bar{x}_2 x_3$$

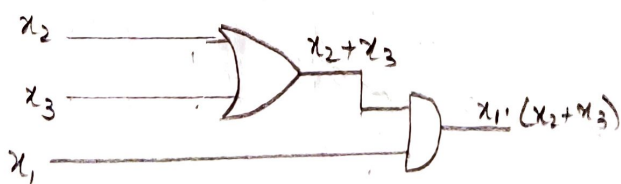
$$= x_1 \bar{x}_2 x_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 x_3$$

$$= x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_3 = x_1 x_3 (x_2 + \bar{x}_2) + x_1 x_2 \bar{x}_3$$

$$= x_1 x_3 + x_1 x_2 \bar{x}_3 = x_1 (x_3 + x_2 \bar{x}_3)$$

$$= x_1 (x_2 + x_3)$$

$$[y + x] = x + y$$



$$\text{Cost} = 2 + 4 = 6$$

Q Dual & complement of $f = \bar{c}DE$

$$f_D = \bar{c} + \bar{D} + \bar{E} = \bar{c} + D + E$$

$$f' = \overline{\bar{c}DE} = \bar{c} + D + E = c + \bar{D} + \bar{E}$$

$f' = f_D$ complement for each element not to take.

* $f = \bar{x}yz + \bar{x}\bar{y}z$; $w = x\bar{y} \cdot (\bar{x}\bar{y} + xy)$

$f_D = (\bar{x} + y + z) \cdot (\bar{x} + \bar{y} + z)$

$f'_D = \overline{\bar{x}yz + \bar{x}\bar{y}z} = (x + \bar{y} + \bar{z}) \cdot (x + y + \bar{z})$

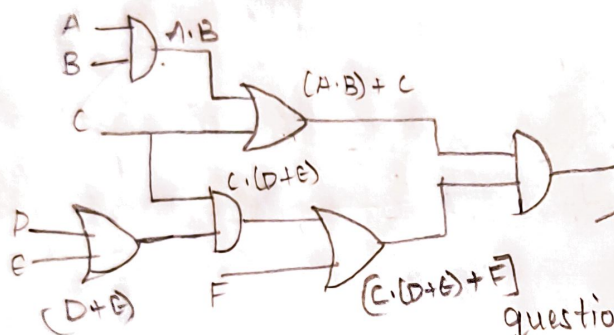
$W_D = (x + \bar{y}) + [(\bar{x} + \bar{y}) \cdot (x + y)]$

$W' = \overline{x\bar{y} \cdot (\bar{x}\bar{y} + xy)} = \overline{x\bar{y}} + (\overline{\bar{x}\bar{y}} + \overline{xy}) = (\bar{x} + y) + [(x + \bar{y})]$

only variables are complemented b/w f_D & f' (or)

W_D & W'

* Convert to NAND & NOR



Multi level circuit

logical expression

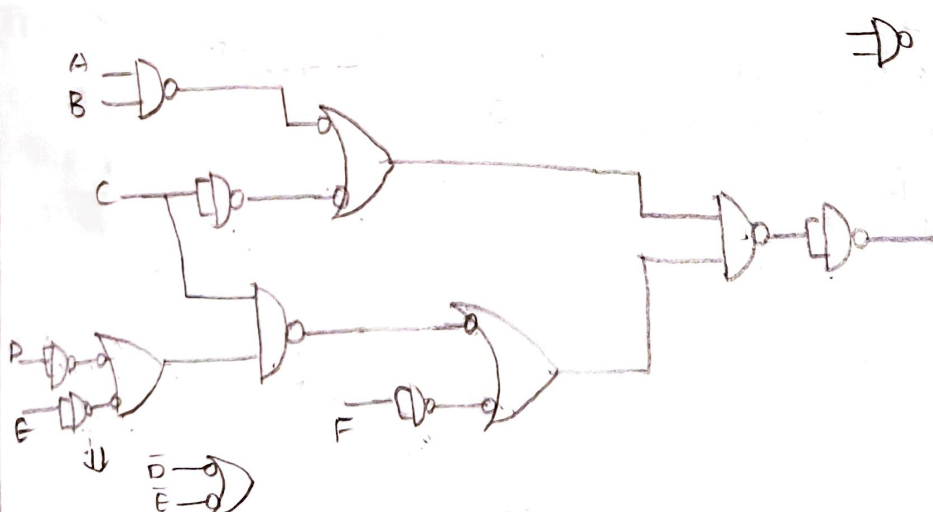
0 & 1 → output

convert NAND / NOR gates

questions on it

$[(A \cdot B) + C] \cdot [C \cdot (D + E) + F]$

Convert to NAND



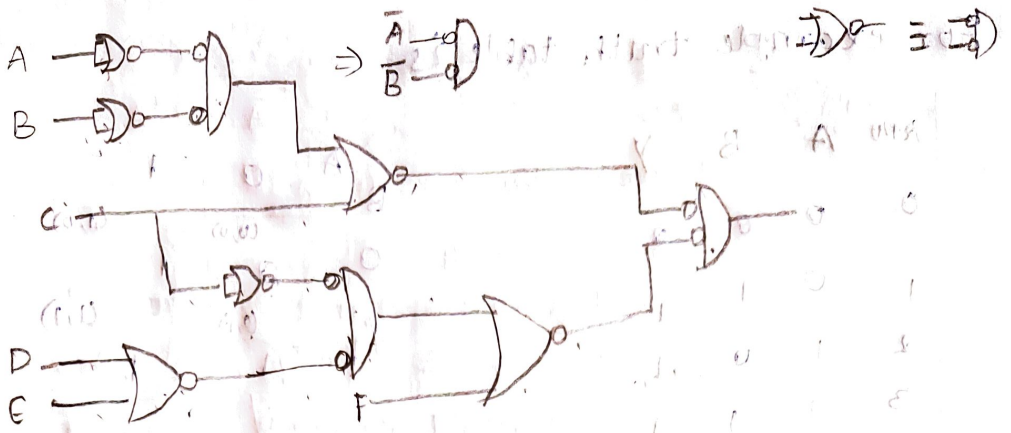
$\Rightarrow \text{NAND} \equiv \Rightarrow \text{NAND}$

→ If truth table can be expressed in high output it is equally correct when it is expressed in low output.

$5 + 5 + 5 = 15$

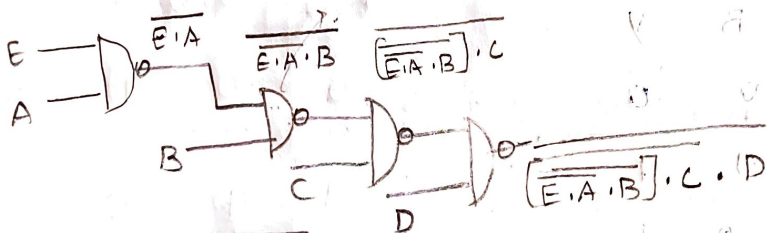
But of you know...

convert to NOR



cost = 9 + 16 = 25

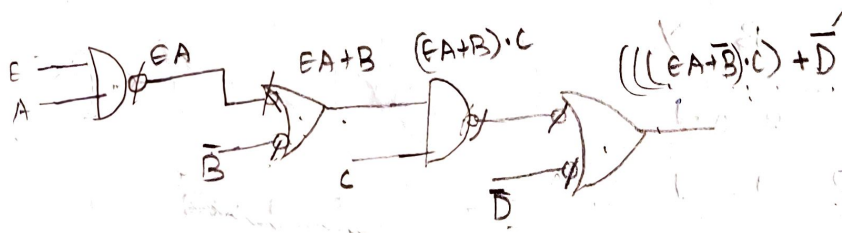
Q write an expression for circuit.



$$= \overline{\overline{E.A.B.C.D}} = \overline{\overline{E.A.B.C} + \overline{D}} = (\overline{\overline{E.A.B}})C + \overline{D}$$

$$= (\overline{E.A + B}).C + \overline{D} = ((E.A) + \overline{B}).C + \overline{D}$$

→ equal.



★ Karnaugh Map (K-Map)

For example truth table is

RNO	MSB LSB		Y
	A	B	
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

Group adjacent one's

RNO	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

RNO	A	B	Y
0	0	0	1
1	0	1	1
2	1	0	1
3	1	1	1

	A \ B	$\bar{A}$	A
		0	1
B	0	0	0
1	0	0	1
1	1	1	1
0	1	1	0

$$f = A + B$$

	A \ B	$\bar{A}$	A
		0	1
B	0	0	0
1	0	0	1
1	1	1	0
0	1	0	0

$$f = A'B + AB' = A \oplus B$$

	A \ B	$\bar{A}$	A
		0	1
B	0	1	1
1	0	1	1
1	1	1	1
0	1	1	1

$f = 1$ because it doesn't depend on A & B

2^n can be in a group $n=0,1,2,3, \dots$

3 variable. K-Map

$\Sigma m(1,2,3,5,6)$

$$XY + YZ + XZ$$

A_2	A_1	A_0	Y_1
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$A_2 A_1$	$\bar{A}_2 \bar{A}_1$	$\bar{A}_2 A_1$	$A_2 \bar{A}_1$	$A_2 A_1$
A_0	00	01	11	10
$\bar{A}_0$	0	0	1	1
A_0	1	1	0	0

4PI
2EPI's

$$Y_1 = \bar{A}_2 \bar{A}_0 + \bar{A}_2 A_1 + \bar{A}_1 \bar{A}_0 + A_1 \bar{A}_0$$

minimal cover = 3.

$$= \bar{A}_2 A_1 + \bar{A}_1 A_0 + A_1 \bar{A}_0$$

$A_2 A_1$	$\bar{A}_2 \bar{A}_1$	$\bar{A}_2 A_1$	$A_2 \bar{A}_1$	$A_2 A_1$
A_0	00	01	11	10
$\bar{A}_0$	0	0	1	1
A_0	1	1	1	1

(1,2,3,5,6,7)

2 quads.

1 group $\rightarrow$ 3 var product
2 group $\rightarrow$ 2 var product
4 group $\rightarrow$ 1 var (Y_2 ex)

2PI
2EPI's

$$Y_2 = \bar{A}_0 + A_1$$

$A_2 A_1$	$\bar{A}_2 \bar{A}_1$	$\bar{A}_2 A_1$	$A_2 \bar{A}_1$	$A_2 A_1$
A_0	00	01	11	10
$\bar{A}_0$	1	1	0	0
A_0	1	1	0	0

(0,1,2,3,4,5)

2PI
2EPI's

$$Y_3 = \bar{A}_1 + \bar{A}_2$$



4 Variable K-Map → Rolling property → left to right, top to bottom

MSB A ₃ A ₂ LSB A ₁ A ₀	$\overline{A_3}\overline{A_2}$ 00	$\overline{A_3}A_2$ 01	$A_3\overline{A_2}$ 11	A_3A_2 10
$\overline{A_1}\overline{A_0}$ 00	(0000) 0 0	(0100) 4 0	(1100) 12 1	(1000) 8 0
$\overline{A_1}A_0$ 01	(0001) 1 0	(0101) 5 0	(1101) 13 1	(1001) 9 1
$A_1\overline{A_0}$ 11	(0011) 3 1	(0111) 7 1	(1111) 15 1	(1011) 11 0
A_1A_0 10	(0010) 2 0	(0110) 6 0	(1110) 14 1	(1010) 10 0

$f(A_3, A_2, A_1, A_0) = \sum m(3, 7, 9, 12, 13, 14, 15)$

→ All product terms are called as prime implicants

→ 1 group → 4 variable product term

- 2 " → 3 " "
- 4 " → 2 " "
- 8 " → 1 " "

4 prime implicants
↓
each one is literal

MSB A ₃ A ₂ LSB A ₁ A ₀	$\overline{A_3}\overline{A_2}$ 00	$\overline{A_3}A_2$ 01	$A_3\overline{A_2}$ 11	A_3A_2 10
$\overline{A_1}\overline{A_0}$ 00	* (1) 0	0 4	0 12	* (1) 8
$\overline{A_1}A_0$ 01	0 1	0 5	0 13	0 9
$A_1\overline{A_0}$ 11	* 1 3	1 7	1 15	0 11
A_1A_0 10	1 2	1 6	0 14	* 10

$f(A_3, A_2, A_1, A_0) = \sum M(0, 2, 3, 6, 7, 8, 10, 15)$

3 PI
 3 EPI's

→ Each variable of a prime implicant is literal.

EPI → essential prime implicant

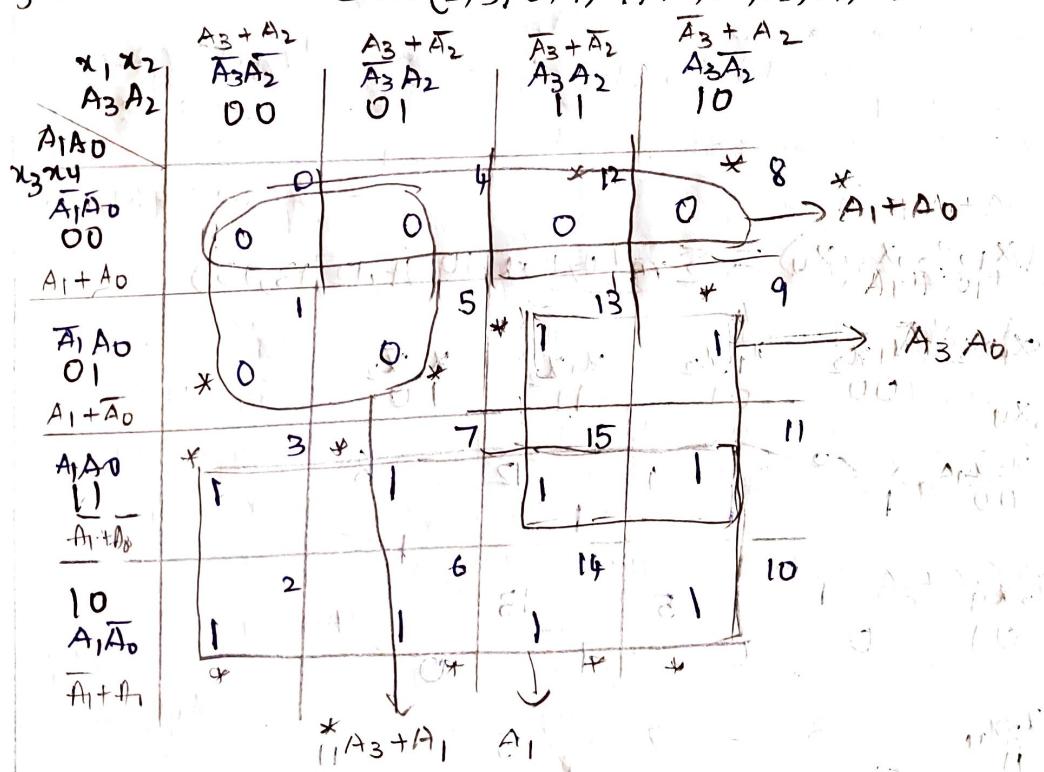
EPI $\Rightarrow A_1 A_0 \bar{A}_3 \rightarrow \textcircled{1}$
 $A_3 A_2 \rightarrow \textcircled{2}$
 $\bar{A}_1 A_0 A_3 \rightarrow \textcircled{3}$ 3 EPI's

$f(A_3, A_2, A_1, A_0) = A_1 A_0 \bar{A}_3 + A_3 A_2 + \bar{A}_1 A_0 A_3$ → minimal cover / minimal SOP

$f(A_3, A_2, A_1, A_0) = \sum m(0, 2, 3, 6, 7, 8, 10, 15)$
 $= A_1 \bar{A}_3 + \bar{A}_0 \bar{A}_2 + A_1 A_0 A_2$

if 14 is also 1 then 4 PI's occur (2, 6, 14, 10) → PI
 but 3 EPI's will only be there.

Q1. $f(x_1, x_2, x_3, x_4) = \sum m(2, 3, 6, 7, 9, 10, 11, 13, 14, 15)$



$Y = A_1 + A_0 A_3 = x_3 + x_4 x_1 \rightarrow$ Minimal SOP expression

2 EPI's

2 PI's

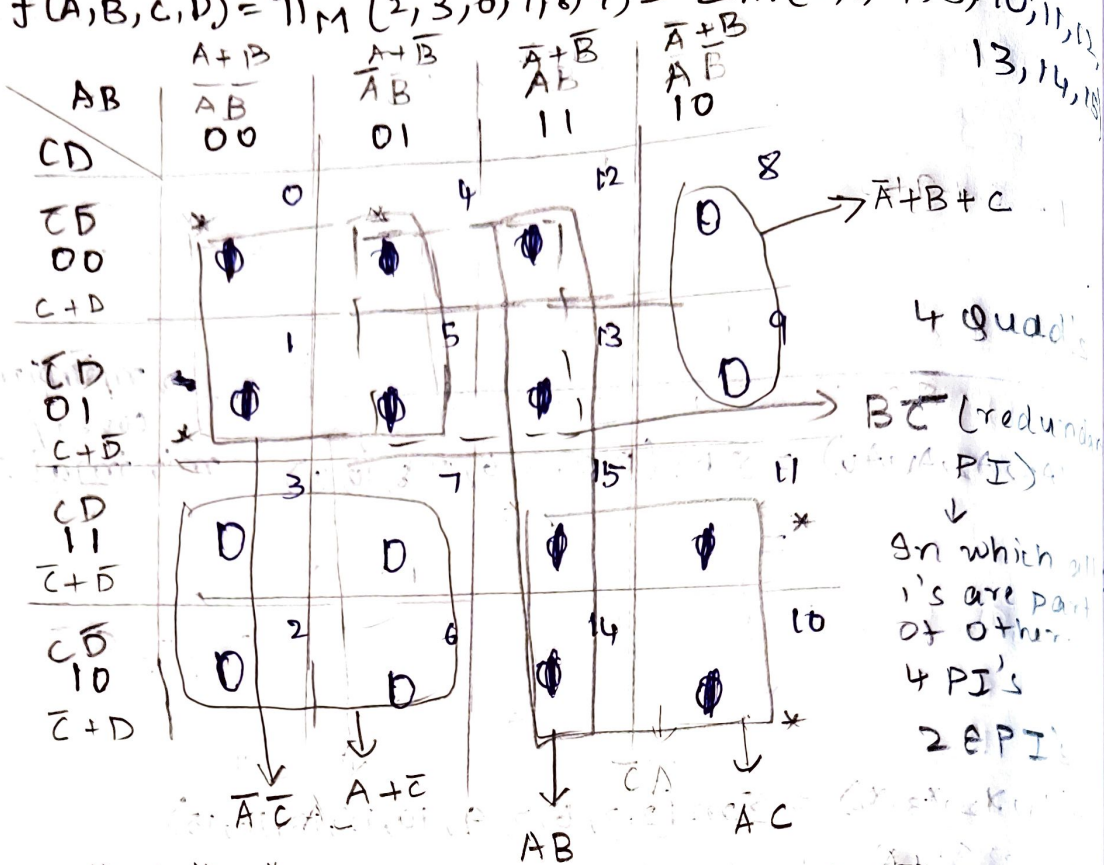
$Y = (A_1 + A_0)(A_3 + A_1) = (x_3 + x_4)(x_1 + x_3)$

$= (A_1 A_3) + A_0 A_1 + A_0 A_3 + A_0 A_1$

$= A_1 A_3 + A_1 + A_0 A_3 + A_0 A_1$

$= A_1 + A_1 A_0 + A_0 A_3 = A_1 + A_0 A_3$

Q2) $f(A, B, C, D) = \Pi_M(2, 3, 6, 7, 8, 9) = \sum m(0, 1, 4, 5, 10, 11, 12, 13, 14, 15)$



$$Y = \bar{A}\bar{C} + AC + AB + BC \rightarrow \text{Min SOP exp}$$

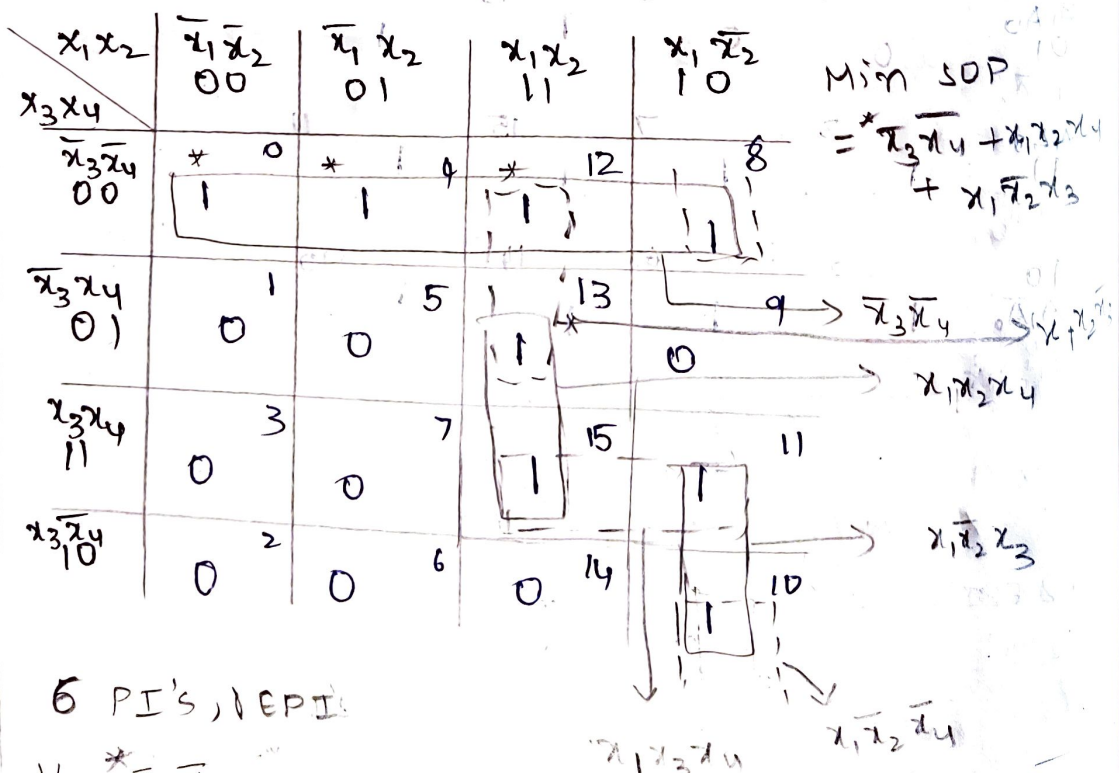
$$Y = \bar{A}\bar{C} + AC + AB + BC$$

$$= \bar{A}\bar{C} + AC + AB + \bar{A}\bar{B} + \bar{A}B + AB + BC$$

$$= \bar{A}\bar{C} + AC + AB + BC$$

4 PI's

Q3) $f(x_1, x_2, x_3, x_4) = \sum m(0, 4, 8, 10, 11, 12, 13, 15)$



$$Y = \bar{x}_3\bar{x}_4 + x_1x_2x_4 + x_1\bar{x}_2x_3 + \bar{x}_3\bar{x}_4 + x_1x_2x_4 + x_1\bar{x}_2x_3 + x_1x_2x_4 + x_1\bar{x}_2x_3$$

Q4) $f(x_1, x_2, x_3, x_4) = \sum m(0, 2, 4, 5, 10, 11, 13, 15)$

$x_3 x_4$ \ $x_1 x_2$	$\bar{x}_1 \bar{x}_2$ 00	$\bar{x}_1 x_2$ 01	$x_1 \bar{x}_2$ 11	$x_1 x_2$ 10
$\bar{x}_3 \bar{x}_4$ 00	0	1	1	0
$\bar{x}_3 x_4$ 01	0	1	1	0
$x_3 \bar{x}_4$ 11	0	0	1	1
$x_3 x_4$ 10	1	0	0	1

$\bar{x}_1 \bar{x}_3 \bar{x}_4$ (from $\bar{x}_3 \bar{x}_4$ row, $\bar{x}_1 \bar{x}_2$ columns)
 $x_2 \bar{x}_3 x_4$ (from $\bar{x}_3 x_4$ row, $\bar{x}_1 x_2$ columns)
 $x_1 x_3 x_4$ (from $x_3 \bar{x}_4$ row, $x_1 x_2$ columns)
 $x_1 \bar{x}_2 x_3$ (from $x_3 x_4$ row, $\bar{x}_1 \bar{x}_2$ columns)

8 PI's

NO EPI's

$$Y = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_2 \bar{x}_3 x_4 + x_1 x_3 x_4 + x_1 \bar{x}_2 x_3$$

Minimal SOP = $\bar{x}_1 \bar{x}_3 \bar{x}_4 + x_2 \bar{x}_3 x_4 + x_1 x_3 x_4 + x_1 \bar{x}_2 x_3$

Q5) $f(A_3, A_2, A_1, A_0) = \sum m(3, 4, 5, 7, 9, 13, 14, 15)$

$A_3 A_2$ \ $A_1 A_0$	$\bar{A}_3 \bar{A}_2$ 00	$\bar{A}_3 A_2$ 01	$A_3 \bar{A}_2$ 11	$A_3 A_2$ 10
00 $\bar{A}_1 \bar{A}_0$	0	1	0	0
01 $\bar{A}_1 A_0$	0	1	1	1
11 $A_1 \bar{A}_0$	1	1	1	0
10 $A_1 A_0$	0	0	1	0

$\bar{A}_3 A_2 \bar{A}_1$ (from $\bar{A}_1 \bar{A}_0$ row, $\bar{A}_3 A_2$ columns)
 $A_3 \bar{A}_1 A_0$ (from $\bar{A}_1 A_0$ row, $A_3 A_2$ columns)
 $\bar{A}_3 A_1 A_0$ (from $A_1 \bar{A}_0$ row, $\bar{A}_3 A_2$ columns)
 $A_2 A_0$ (from $A_1 A_0$ row, $\bar{A}_3 A_2$ columns)
 $A_3 A_2 A_1$ (from $A_1 A_0$ row, $A_3 A_2$ columns)

5 PI's

4 EPI's

$$Y = \bar{A}_3 A_1 A_0 + A_3 \bar{A}_2 A_1 + \bar{A}_3 \bar{A}_1 A_0 + \bar{A}_2 A_2 \bar{A}_1 \rightarrow \text{Minimal SOP}$$

* Literal :- A single variable or it's complement.

★ Step for Finding minimal cover.

1. Generate all P.I for given function f .
2. Find E.P.I's
3. E.P.I's are part of answer + other P.I (if any of the 1 is left in k-Map) cover that using minimal expression.

★ 5-Variable k MAP

RNO	A	B	C	D	E
0	0	0	0	0	0
1	0	0	0	0	1
2	0	0	0	1	0
3	0	0	0	1	1
4	0	0	1	0	0
5	0	0	1	0	1
6	0	0	1	1	0
7	0	0	1	1	1
8	0	1	0	0	0
9	0	1	0	0	1
10	0	1	0	1	0
11	0	1	0	1	1
12	0	1	1	0	0
13	0	1	1	0	1
14	0	1	1	1	0
15	0	1	1	1	1
16	1	0	0	0	0
17	1	0	0	0	1
18	1	0	0	1	0
19	1	0	0	1	1
20	1	0	1	0	0
21	1	0	1	0	1
22	1	0	1	1	0
23	1	0	1	1	1
24	1	1	0	0	0

RNO	A	B	C	D	E
25	1	1	0	0	1
26	1	1	0	1	0
27	1	1	0	1	1
28	1	1	1	0	0
29	1	1	1	0	1
30	1	1	1	1	0
31	1	1	1	1	1

A=0

BC \ DE	00	01	11	10
00	1	1	0	0
01	0	0	1	1
11	0	0	0	0
10	1	1	0	0

$\overline{A}\overline{B}\overline{E}$

A=1

BC \ DE	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	0	1	1	0
10	0	0	0	0

ACE

- * Single - 5 var product term
- pair - 4 var " "
- quad - 3 var " "
- octat - 2 var " "
- 16 - 1 var " "

$$f = \sum m(0, 2, 4, 6, 9, 13, 21, 23, 25, 29, 31)$$

= $\overline{A}\overline{B}\overline{E} + ACE + B\overline{D}E$ → Minimal S.O.P expression.

P.I=3, E.P.I=3, All three are part of answer.

$$Q. f = \sum m(0, 1, 2, 3, 7, 9, 11, 13, 15, 21, 23, 24, 25, 26, 27, 29, 31)$$

A=0

A=1

BC \ DE	00	01	11	10
00	1	0	0	0
01	1	0	1	1
11	1	1	1	1
10	1	0	0	0

$\overline{A}\overline{B}\overline{C}$ $\overline{A}DE$

BC \ DE	00	01	11	10
00	0	0	0	1
01	0	1	1	1
11	0	1	1	1
10	0	0	0	1

BE ACE $\overline{A}\overline{B}\overline{C}$

$$= \overline{A}\overline{B}\overline{C} + \overline{A}\overline{B}\overline{C} + BE + ACE + \overline{A}DE$$

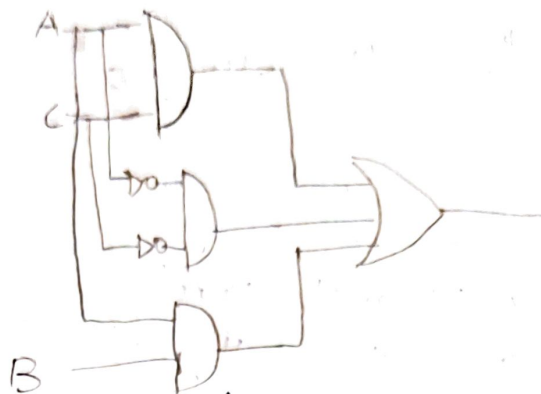
6 P.I's
4 E.P.I's
5 Minimal.

8. $f = \pi M(2, 3, 6, 7, 8, 9)$

S.O.P = $\bar{A}\bar{E} + AC + AB$

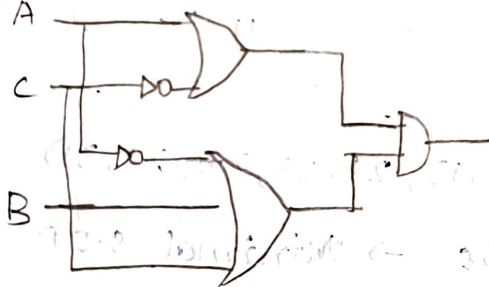
P.O.S = $(A+E)(\bar{A}+B+C)$

S.O.P =



Cost = $4 + 9$
= 13

P.O.S = A



Cost = $3 + 7$
= 10

∴ Pos is better than s.o.p

9. $f = \Sigma m(0, 1, 2, 3, 7, 9, 11, 13, 15, 21, 23, 24, 25, 26, 27, 29, 31)$

$= \pi M(4, 5, 6, 8, 10, 12, 14, 16, 17, 18, 19, 20, 22, 28, 30)$

A=0				
BC	B+C	B+C	$\bar{B}+\bar{C}$	$\bar{B}+C$
DE	00	01	10	11
D+E				
00	0	4	12	8
01	1	5	13	9
10	2	6	14	10
11	3	7	15	11

A=1				
BC	B+C	B+C	$\bar{B}+\bar{C}$	$\bar{B}+C$
DE	00	01	10	11
D+E				
00	16	20	28	24
01	17	21	29	25
10	18	22	30	26
11	19	23	31	27

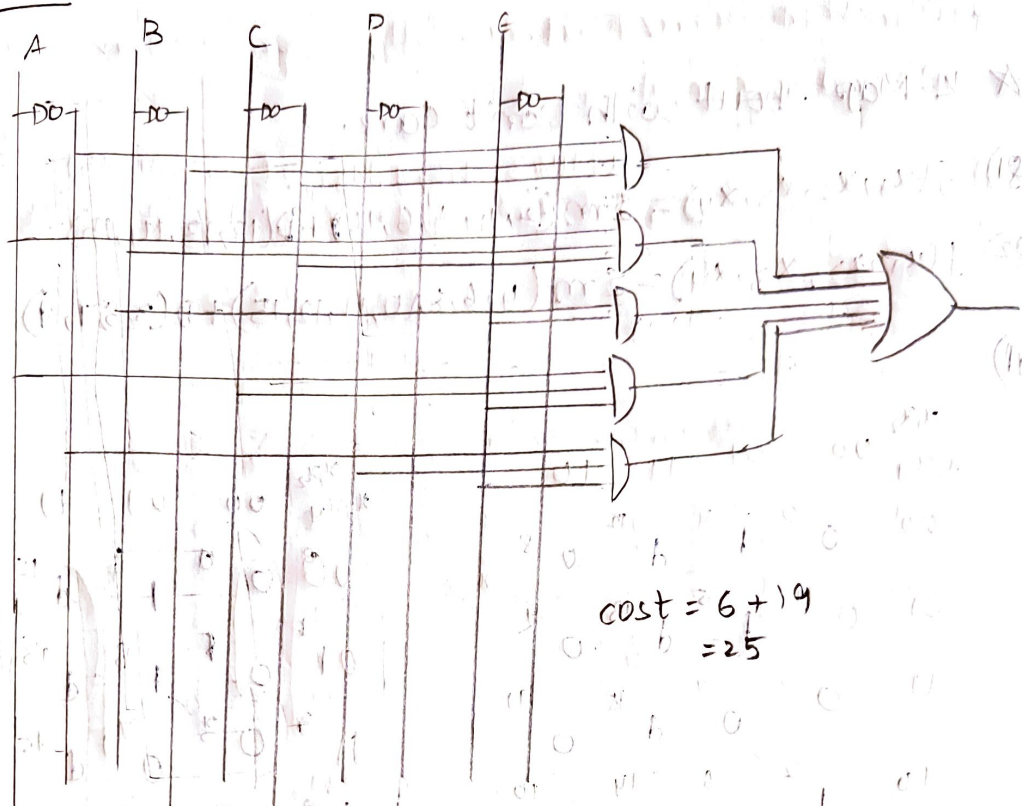
$A+B+\bar{C}+D$

$A+\bar{B}+E$

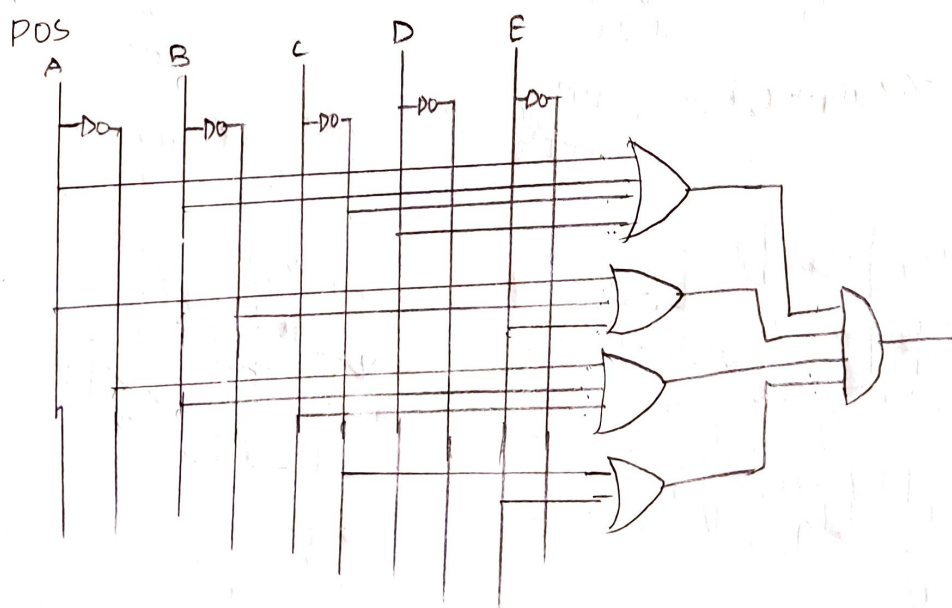
$\bar{A}+B+C$

$$= (A+B+C+D)(A+\bar{B}+E)(\bar{A}+B+C)(\bar{C}+E)$$

$$\underline{SOP} = \bar{A}\bar{B}\bar{C} + AB\bar{C} + BE + ACE + \bar{A}DE$$



$$\text{cost} = 6 + 19 = 25$$



$$\text{cost} = 5 + 16 = 21$$

∴ POS is better than SOP

BED. binary code decimal (0-9)

→ Don't care → d/x/-

↳ when the inputs are not known we have write

★ K-Map about with Don't care.

$$= \pi M(0, 3, 7, 8, 9, 11), D(12, 13, 14, 15)$$

Q1) $f(x_1, x_2, x_3, x_4) = \sum m(2, 4, 5, 6, 10) + D(12, 13, 14, 15)$

Q2) $f(x_1, x_2, x_3, x_4) = \sum m(4, 6, 8, 10, 11, 12, 15) + D(3, 5, 7, 9)$

SOP

POS

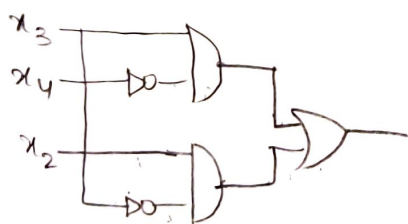
Q1)

x_1x_2	00	01	11	10
x_3x_4	00	01	11	10
x_3x_4	00	01	11	10
x_3x_4	01	01	11	10
x_3x_4	11	01	11	10
x_3x_4	10	01	11	10

Groupings for SOP:
 - Group 1: $x_3\bar{x}_4$ (cells 2, 4, 6, 10)
 - Group 2: $x_2\bar{x}_3$ (cells 4, 5, 12, 13)

$$= x_3\bar{x}_4 + x_2\bar{x}_3$$

3 PI's
2 EPI's



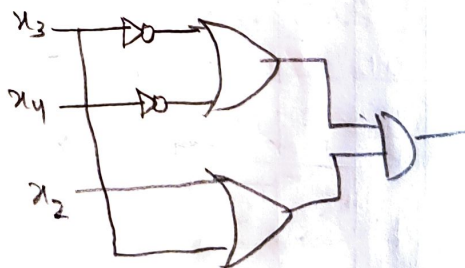
$$\text{Cost} = 3 + 6 = 9$$

∴ SOP & POS are same.

x_1x_2	00	01	11	10
x_3x_4	00	01	11	10
x_3x_4	00	01	11	10
x_3x_4	01	01	11	10
x_3x_4	11	01	11	10
x_3x_4	10	01	11	10

Groupings for POS:
 - Group 1: $(\bar{x}_3 + \bar{x}_4)$ (cells 0, 1, 2, 3)
 - Group 2: $(x_2 + x_3)$ (cells 4, 5, 6, 7)

$$= (\bar{x}_3 + \bar{x}_4) \cdot (x_2 + x_3)$$



$$\text{Cost} = 3 + 6 = 9$$

Q2) Cost is same as Q1

92)

$x_3 x_4$	$x_1 x_2$ 00	$\bar{x}_1 x_2$ 01	$x_1 \bar{x}_2$ 11	$x_1 x_2$ 10
$\bar{x}_3 \bar{x}_4$ 00	0	1	1	8
$\bar{x}_3 x_4$ 01	0	d	0	d
$x_3 \bar{x}_4$ 11	d	d	1	1
$x_3 x_4$ 10	0	2	0	10

$\bar{x}_1 x_2$
 $x_3 x_4$
 $x_1 \bar{x}_2$

$x_3 x_4$	$x_1 + x_2$ 00	$x_1 + \bar{x}_2$ 01	$\bar{x}_1 + x_2$ 11	$\bar{x}_1 + \bar{x}_2$ 10
$\bar{x}_3 \bar{x}_4$ 00	0	1	1	1
$\bar{x}_3 x_4$ 01	0	d	0	d
$\bar{x}_3 + \bar{x}_4$ 11	d	d	1	1
$\bar{x}_3 + x_4$ 10	0	1	0	1

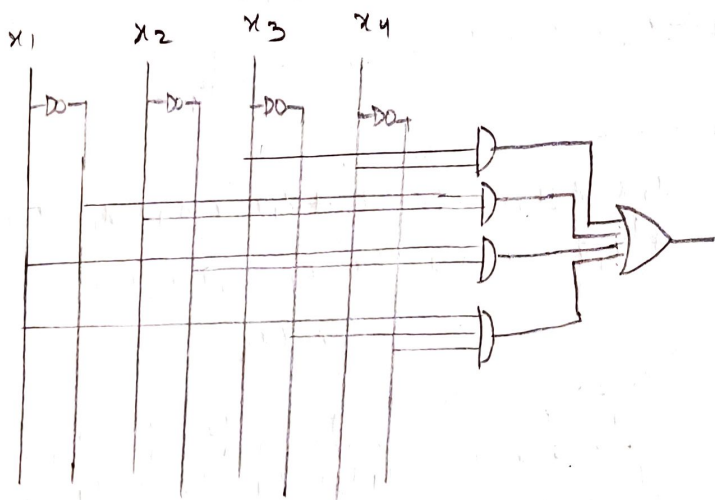
$x_1 + x_2$
 $x_3 + \bar{x}_4$
 $\bar{x}_3 + \bar{x}_4$
 $\bar{x}_3 + x_4$

SPI'S
 BEPI'S

BEPI'S

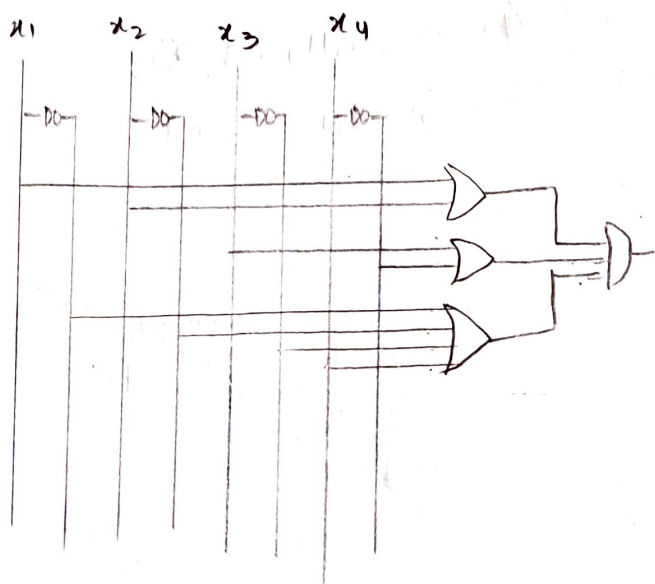
$$(x_1 + x_2) \cdot (x_3 + \bar{x}_4) \cdot (\bar{x}_1 + \bar{x}_2 + \bar{x}_3 + x_4)$$

$$x_3 x_4 + \bar{x}_1 x_2 + x_1 \bar{x}_2 + x_1 \bar{x}_3 \bar{x}_4$$



SOP

$$\text{cost} = 5 + 13 = 18$$



POS

$$\text{cost} = 4 + 11 = 15$$

∴ Pos is better than SOP

Q. $f(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 \bar{x}_3 x_4$

Assume $D = \Sigma(9, 12, 14)$

i) Find Minterm, Maxterm list

ii) Find Minimal SOP & POS.

1) $f(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 \bar{x}_3 x_4$

$$\bar{x}_1 \bar{x}_3 \bar{x}_4 = \bar{x}_1 \cdot 1 \cdot \bar{x}_3 \cdot \bar{x}_4 = \bar{x}_1 \cdot (x_2 + \bar{x}_2) \cdot \bar{x}_3 \cdot \bar{x}_4$$

$$= \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4$$

$$x_3 x_4 = 1 \cdot x_3 x_4 = (x_2 + \bar{x}_2) x_3 x_4 = x_2 x_3 x_4 + \bar{x}_2 x_3 x_4$$

$$= 1 \cdot x_2 x_3 x_4 + 1 \cdot \bar{x}_2 x_3 x_4 = (x_1 + \bar{x}_1) x_2 x_3 x_4 + (x_1 + \bar{x}_1) \bar{x}_2 x_3 x_4$$

$$= x_1 x_2 x_3 x_4 + \bar{x}_1 x_2 x_3 x_4 + x_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 \bar{x}_2 x_3 x_4$$

$$x_1 \bar{x}_2 x_4 = \bar{x}_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4$$

$$x_1 x_2 \bar{x}_3 x_4$$

$$f(x_1, x_2, x_3, x_4) = \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 + x_1 x_2 x_3 x_4 + \bar{x}_1 x_2 x_3 x_4$$

$$+ x_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 x_2 \bar{x}_3 x_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4$$

$$+ x_1 x_2 \bar{x}_3 x_4$$

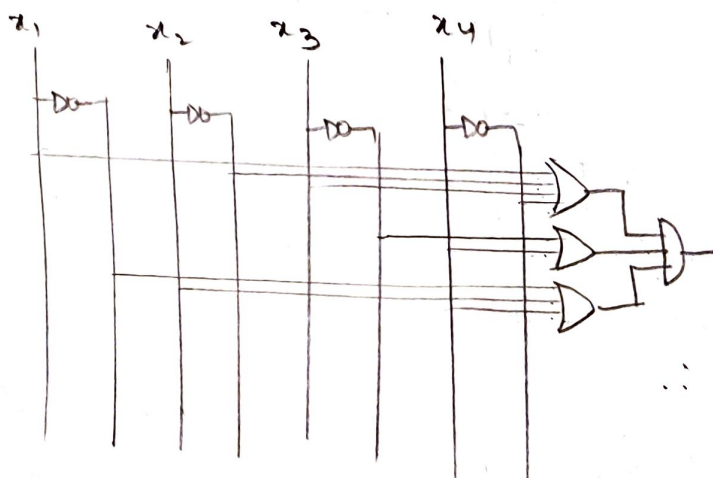
$$= \bar{x}_1 x_2 \bar{x}_3 \bar{x}_4 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \bar{x}_4 + x_1 x_2 x_3 x_4 + \bar{x}_1 x_2 x_3 x_4$$

$$+ x_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 \bar{x}_2 x_3 x_4 + \bar{x}_1 x_2 \bar{x}_3 x_4 + x_1 x_2 \bar{x}_3 x_4$$

$$= \Sigma m(4, 0, 15, 7, 11, 3, 1, 13)$$

$$= \Pi M(1, 2, 5, 6, 8, 10, 15)$$

POS



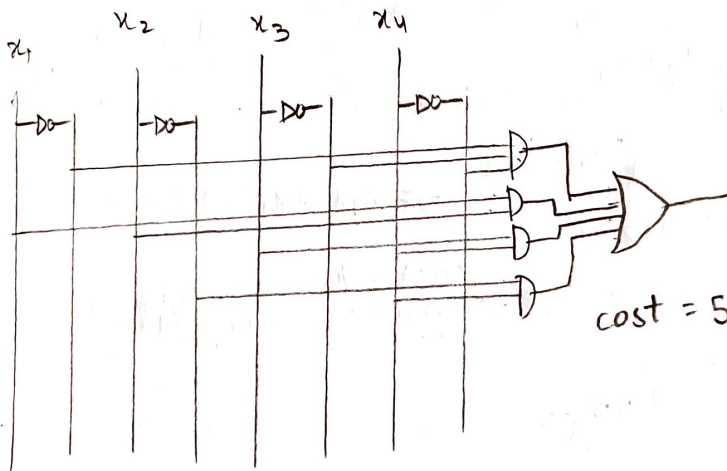
$$\text{cost} = 4 + 2 = 16$$

$\therefore$ POS is better.

$x_1 x_2$	$\bar{x}_1 \bar{x}_2$ 00	$\bar{x}_1 x_2$ 01	$x_1 \bar{x}_2$ 11	$x_1 x_2$ 10
$x_3 x_4$				
00 $\bar{x}_3 \bar{x}_4$	1	1	d	0
01 $\bar{x}_3 x_4$	1	0	1	d
11 $x_3 x_4$	1	1	1	1
10 $\bar{x}_3 x_4$	0	0	d	0

$\bar{x}_1 \bar{x}_3 \bar{x}_4$ $x_1 x_2$ $\bar{x}_2 x_4$ $x_3 x_4$

$$SOP = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_1 x_2 + x_3 x_4 + \bar{x}_2 x_4$$



$$\text{cost} = 5 + 13 = 18$$

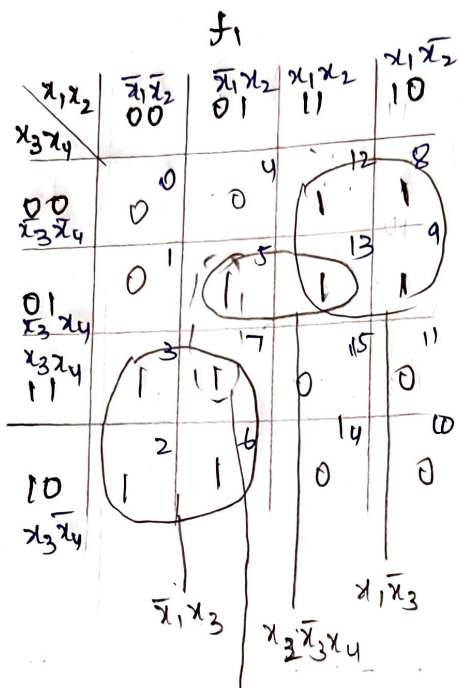
$x_1 x_2$	$x_1 + x_2$ 00	$\bar{x}_1 + \bar{x}_2$ 01	$\bar{x}_1 + x_2$ 11	$\bar{x}_1 + x_2$ 10
$x_3 x_4$				
00 $\bar{x}_3 + \bar{x}_4$	1	1	d	0
01 $\bar{x}_3 + x_4$	1	0	1	d
11 $x_3 + \bar{x}_4$	1	1	1	1
10 $x_3 + x_4$	0	0	d	0

$\bar{x}_1 + x_3 + x_4$
 $x_1 + \bar{x}_2 + x_3 + \bar{x}_4$
 $(x_1 + \bar{x}_2 + x_3 + \bar{x}_4) \cdot (\bar{x}_3 + x_4)$
 $(\bar{x}_1 + x_2 + x_4)$
 $\bar{x}_3 + x_4$ $\bar{x}_1 + x_2 + x_4$

★ Multioutput circuit.

Q. $f_1(x_1, x_2, x_3, x_4) = x_1 \bar{x}_3 + \bar{x}_1 x_3 + x_2 \bar{x}_3 x_4$

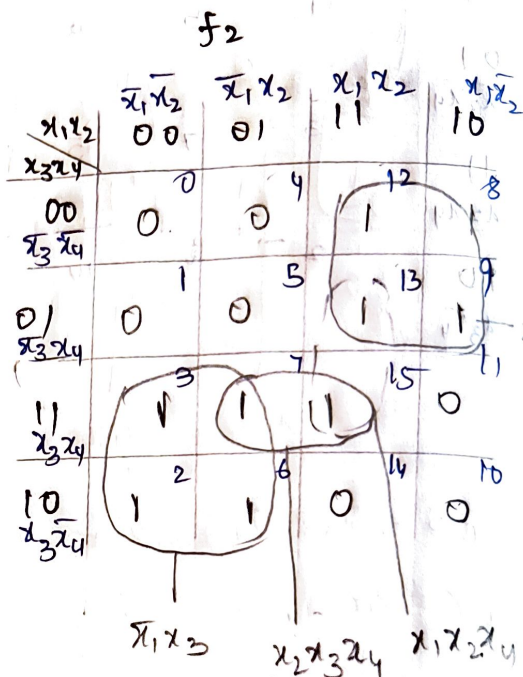
$f_2(x_1, x_2, x_3, x_4) = x_1 \bar{x}_3 + \bar{x}_1 x_3 + x_2 x_3 x_4$



$\bar{x}_1 x_2 x_4$

$f = \bar{x}_1 x_3 + x_1 \bar{x}_3 + \bar{x}_1 x_2 x_4$

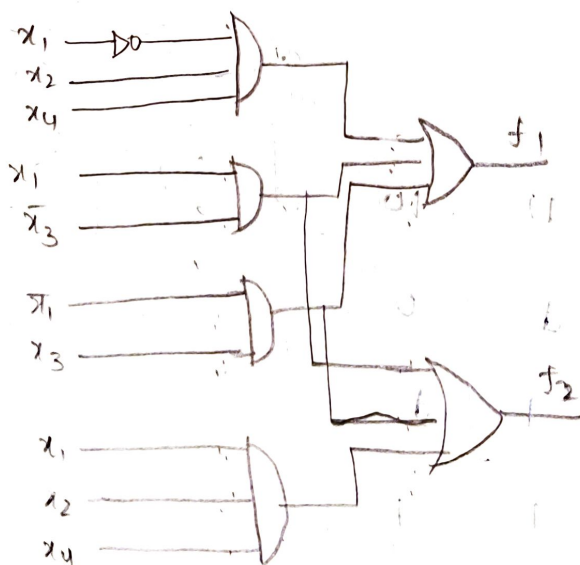
cost = 14



$f = x_1 \bar{x}_3 + \bar{x}_1 x_3 + x_1 x_2 x_4$

cost = 14

Total = 14 + 14 = 28



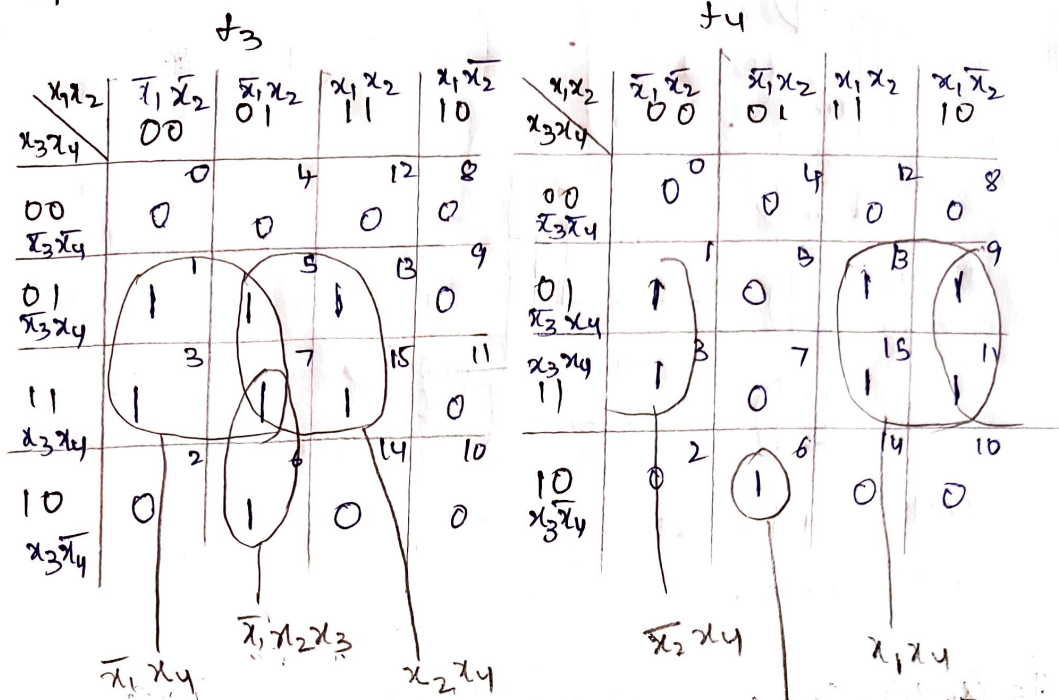
cost = 8 + 4 + 16

= 22

✓ better.

Q. $f_3 = 1, 3, 5, 6, 7, 13, 15 = \Sigma m(1, 3, 5, 6, 7, 13, 15)$

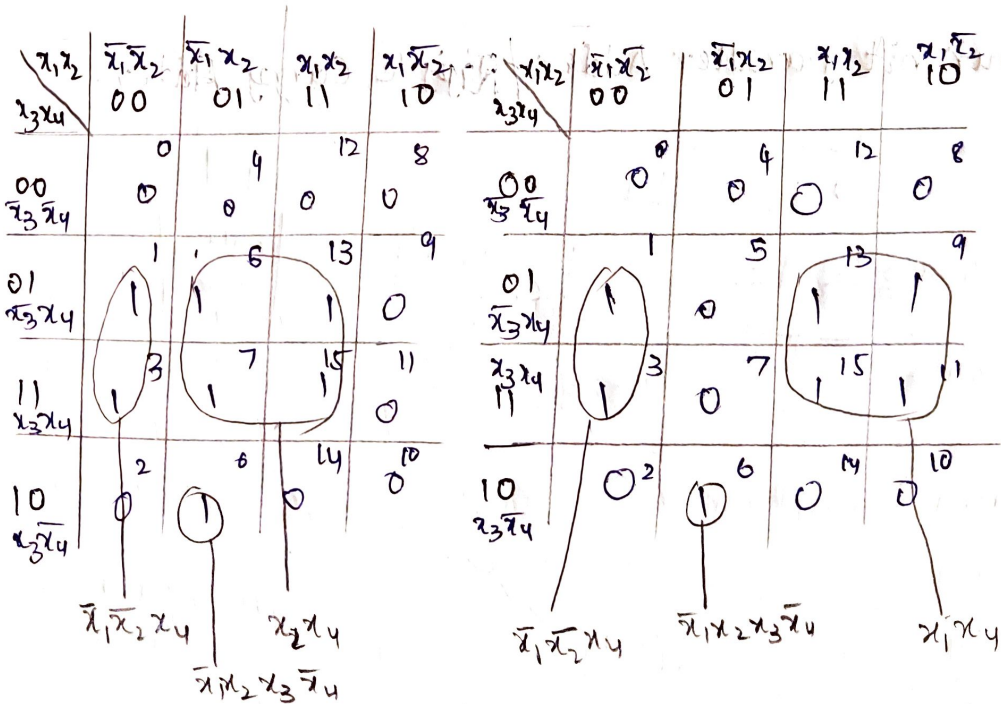
$f_4 = \Sigma m(1, 3, 6, 9, 11, 13, 15)$



$f_3 = \bar{x}_1 x_4 + x_2 x_4 + \bar{x}_1 x_2 x_3$ $f_4 = \bar{x}_2 x_4 + x_1 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$

cost = 14

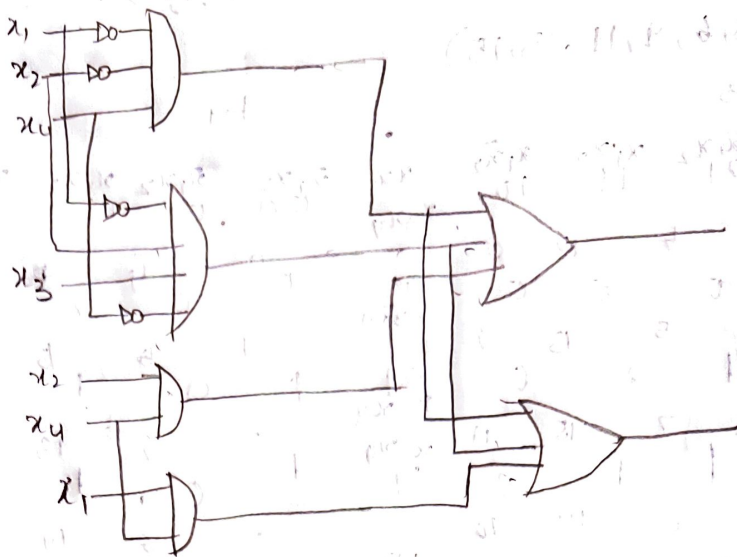
cost = 15 Total = 29.



$f_3 = \bar{x}_1 \bar{x}_2 x_4 + x_2 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$

$f_4 = \bar{x}_1 \bar{x}_2 x_4 + x_1 x_4 + \bar{x}_1 x_2 x_3 \bar{x}_4$

SOP



$$\text{cost} = 6 + 17 = 23$$

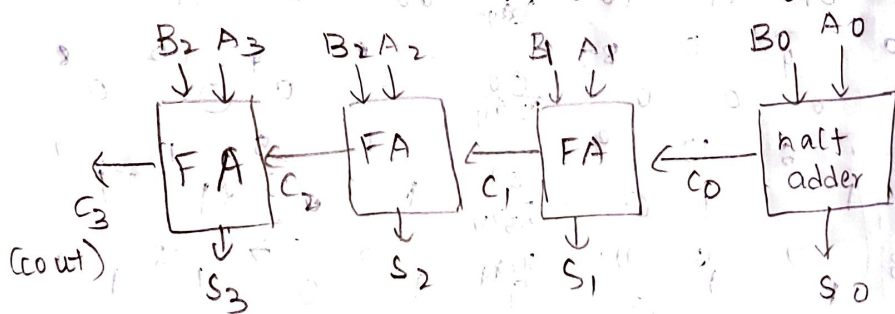
$\therefore$ this is better

Q.

$$f_5(x_1, x_2, x_3) = \sum m(1, 4, 5, 11, 27, 28) + d(0, 12, 14, 15, 20, 31)$$

$$f_6(x_1, x_2, x_3) = \sum m(0, 1, 2, 4, 5, 8, 14, 15, 16, 18, 20, 24, 26, 28, 31) + d(10, 11, 12, 27)$$

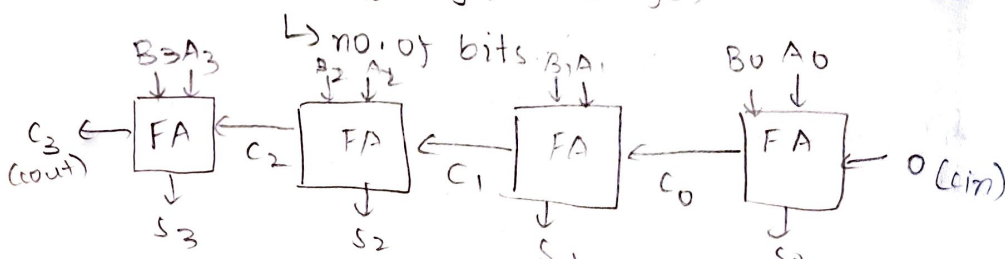
★ Four bit parallel Adder / Ripple Carry Adder.



carry is rippling across stages i.e., one by one adding happens this leads to delay

$$\text{Total Delay} = n \times (\text{Delay for 1 stage})$$

$n = \text{no. of bits / stages}$



Result = $C_3 S_3 S_2 S_1 S_0$

0	1	0	1	C_3	C_2	C_1	C_0	A_3	A_2	A_1	A_0	
0	1	1	0					B_3	B_2	B_1	B_0	
1	1	0	1					C_3	C_2	C_1	C_0	

★ Full Adder

$$S = A \oplus B \oplus C$$

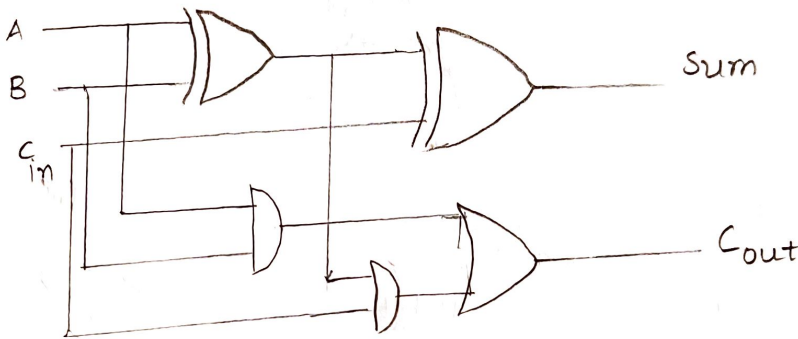
$$C = AB + A C_{in} + B C_{in}$$

$$= AB + ABC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + \bar{A}BC_{in}$$

$$= AB + AB C_{in} + C_{in} (A \oplus B)$$

$$= AB(C_{in} + 1) + C_{in} (A \oplus B)$$

$$= AB + C_{in} (A \oplus B)$$



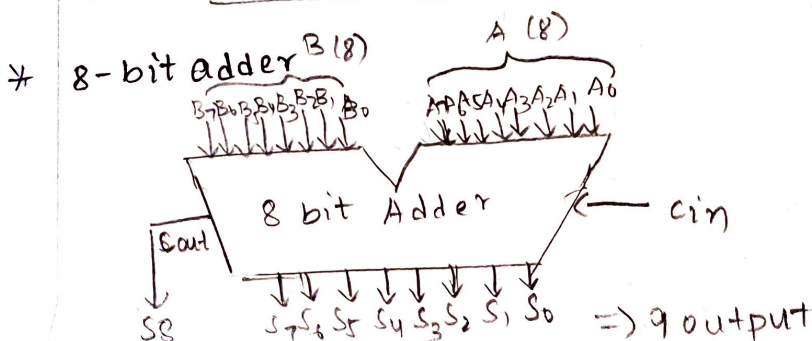
- In a ripple carry Adder which ripples from LSB to MSB.
- It is very slow disadvantage
- advantage — It is simple adder.

Q. $18 - 100110$
 $26 - 110110$

	1	0	0	1	0
18	-	1	0	0	10
26	-	1	1	0	10
	<u>1</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>00</u>
44					

2	26	-0
	<u>13</u>	-1
2	13	-0
	<u>6</u>	-0
2	6	-1
	<u>3</u>	-1
2	3	-1
	<u>1</u>	-1
2	1	-1
	<u>0</u>	-1

2	18	-0
	<u>9</u>	-0
2	9	-0
	<u>4</u>	-0
2	4	-0
	<u>2</u>	-0
2	2	-1
	<u>1</u>	-1
2	1	-1
	<u>0</u>	-1



→

Left shift by 1 = multiply by 2

$N \times 2 \Rightarrow$

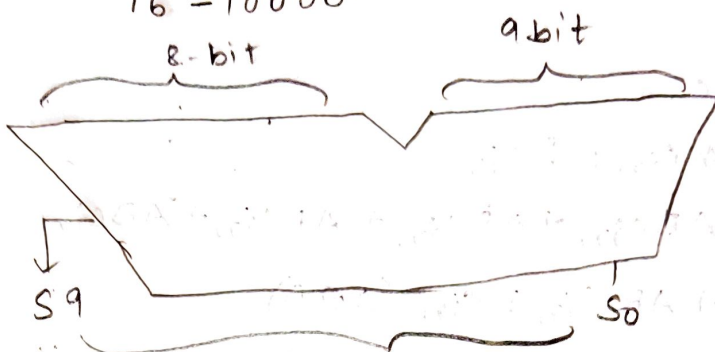
2 - 10
4 - 100
8 - 1000
16 - 10000

5 = 101
10 = 1010
20 = 10100

For 8-bit

$N \times 3 \Rightarrow$

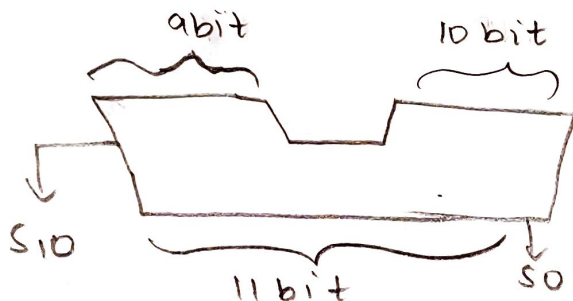
$2N + N$
↓
9 bit 9 bit



$N \times 4 \Rightarrow$ left shift by 2

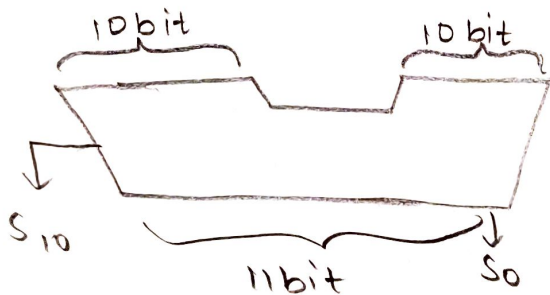
$N \times 5 \Rightarrow$

$3N + 2N$



$6N \Rightarrow$

$3N + 3N$



Q.

26 - 12

$$\begin{array}{r}
 26 \quad - \quad 10 \quad 0 \\
 12 \quad - \quad 1 \quad 1 \quad 0 \quad 0 \\
 \hline
 0 \quad 1 \quad 1 \quad 1 \quad 0 \\
 \hline
 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\
 8 \quad 4 \quad 2 = 14
 \end{array}$$

$$\begin{array}{r}
 26 \\
 12 \\
 \hline
 14
 \end{array}$$

Q.

$$\begin{array}{r}
 0 \quad 1 \quad 1 \quad 2 \\
 10 \quad - \quad 1 \quad 1 \quad 0 \quad 0 \\
 \hline
 0 \quad 0 \quad 1 \quad 1 \quad 0
 \end{array}$$

Decimal	Binary ₂	Octal ₈	Hexa. ₁₆ Decimal	Base-5	BCD
0	0000	0	0 ₁₆	0	0000
1 ₁₀	0001	1 ₈	1	1 ₅	0001
2	0010 ₂	2	2 ₁₆	2 ₅	0010
3	0011	3	3	3	0011 _{BCD}
4	0100 ₂	4 ₈	4	4 ₅	0100
5 ₁₀	0101	5	5 ₁₆	10	0101
6	0110 ₂	6	6	11	0110 _{BCD}
7	0111	7	7	12 ₅	0111
8	1000 ₂	10 ₈	8 ₁₆	13 ₅	1000 _{BCD}
9 ₁₀	1001	11	9	14	1001
10	1010 ₂	12	A	20 ₅	00010000 _{BCD}
11	1011	13 ₈	B ₁₆	21	00010001 _{BCD}
12 ₁₀	1100 ₂	14	C	22 ₅	00010010
13	1101	15	D	23	00010011
14	1110 ₂	16 ₈	E	24	00010100
15 ₁₀	1111 ₂	17	F ₁₆	30 ₅	00010101

Ex:- $10110_{10} = 10110_{10}$

$$10110_2 = 1 \times 2^4 + 1 \times 2^2 + 1 \times 2 = 22_{10}$$

$$10110_8 = 8^4 \times 1 + 8^2 \times 1 + 8^1 \times 1 = 264 \times 64 + 64 + 8 = 4168_{10}$$

$$10110_{16} = 16^4 \times 1 + 16^2 \times 1 + 16 = 65808_{10}$$

→ If binary number has all ones like 1111 then decimal number is $2^n - 1 = 2^5 - 1$

$$15 \times 16^1 + 10 \times 16^0$$

$$FA_{16} = 15 \times 16 + 10 = 250_{10}$$